# Deterministic Semantic Token Compaction, Attention-Sink Structural Anchoring, and Hardware-Aware Pareto Economic Routing for Large Language Model Serving

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***Abstract*—The rapid proliferation of enterprise large language model (LLM) serving and autonomous multi-agent swarms has collided with an operational ceiling: the *LLM Token Economics and Attention Wall*. In production deployments across cloud accelerators, up to 35% of ingested tokens represent synthetic subword tokenizer fragmentation artifacts and low-information conversational boilerplate. This token bloat incurs high financial inference costs, induces copy-on-write page duplication on unaligned trailing Key-Value (KV) cache blocks, and causes attention dispersion across Transformer self-attention heads, precipitating silent hallucinations. In this paper, we present the *Tokenomics Compiler* (`PromptVM`), a deterministic, hardware-grounded compilation framework and discrete-choice economic router for LLM serving. Our system: (1) parses structured schemas into Abstract Syntax Trees (ASTs) and transpiles them into subword-minimal serializations over a bounded candidate space ($|\mathcal{S}_{\text{dense}}| \leq 8$) using quotation-synchronized delta-tokenization while preserving both formal JSON Schema validation contracts and task-semantic parameter role disambiguation; (2) performs sub-clause pleasantry and duplicate-signature span stripping alongside inverse-length-scaled empirical invariant density scoring to prune diffuse conversational verbiage while unconditionally retaining modal and relational invariants; (3) injects uppercase structural landmark anchors with empirical attention affinity boost ($\gamma > 1$), mathematically proving that landmark anchors bound self-attention entropy on dispersive heads ($\mathcal{H}(\mathbf{a}_i) \leq \mathcal{H}_{\max}(i, K, \gamma) < \ln i$ with $\frac{\partial \mathcal{H}_{\max}}{\partial \gamma} < 0$ for all $\gamma > 1$) while co-locating retained constraint predicates to amplify retrieval-head attention by $2.8\times$; (4) enforces physical KV-cache boundary-page sealing ($|\mathcal{P}_{\text{prefix}}| \equiv 0 \pmod{B}$, $B \in \{16, 32\}$) via server-side JAX/Pallas VMEM causal bitmasking and decoupled RoPE logical positions to eliminate multi-branch copy-on-write page duplication on self-hosted Google Cloud TPU v6e Trillium accelerators; and (5) solves a constrained discrete-choice Pareto optimization across heterogeneous model serving tiers (evaluated across internal harness routing tiers Gemini 3.8 Flash, Gemini 3.8 Flash with dynamic test-time reasoning tokens, and Gemini 4.0 Argon, mapped to production Vertex AI Flash and Pro endpoints). Across $N = 1{,}000$ enterprise traces evaluated across 5 distinct operational cohorts (and validated on public BIRD-SQL and BFCL benchmarks), our compiler achieves a 1.11% to 53.47% per-cohort token reduction (17.52% pooled global reduction, saving 81,800 tokens per 1,000 requests; 39.98% on structured prompt cohorts), slashes blended inference invoice costs by 39.44% (up to 88.79% in database queries), and reduces P95 latency by 38.85% against baseline (27.98% against naive regex trimming) while achieving a 99.70% strict first-attempt Turn-1 Pass@1 (vs. 85.00% Turn-1 and 98.50% post-retry on monolithic Argon) and 100.00% post-verification workflow completion rate.**

***Index Terms*—Token Economics, Large Language Models, Transformer Attention Sinks, Subword Compaction, PagedAttention, KV-Cache Alignment, TPU v6e Trillium, Pareto Optimization, Vertex AI, Model Routing.**

Assignee: Google LLC. Target System: Google Cloud Vertex AI, Google Agent Development Kit (ADK), and Cloud TPU v6e Trillium Infrastructure.

## I. Introduction

The industrial transition from basic conversational chat interfaces to high-throughput autonomous multi-agent swarms has fundamentally transformed cloud inference demand. Enterprise platforms such as Google Cloud Vertex AI, the Google Agent Development Kit (ADK; https://cloud.google.com/vertex-ai/docs/agent-builder), and the Model Context Protocol (MCP; https://modelcontextprotocol.io) orchestrate millions of daily reasoning-and-acting invocations [1], [2] wherein software agents query distributed databases, validate OpenAPI endpoints, inspect source repositories, and reconcile financial ledgers.

Despite rapid advancements in accelerator throughput and distributed inference scaling [3], [4] (spanning Google Cloud TPU v4 supercomputing pods [3] and Cloud TPU v6e Trillium accelerators; https://cloud.google.com/tpu/docs/v6e-overview), enterprise deployments face an acute economic and physical bottleneck: the *LLM Token Economics and Attention Wall*. Unmanaged prompt construction introduces severe systemic inefficiencies across three distinct operational layers:

1) **The Subword Fragmentation Tax:** Standard subword tokenizers—operating either via SentencePiece Unigram subword segmentation in Gemini and Gemma [5]–[7] or iterative Byte-Pair Encoding (BPE) pair merges—decompose unaligned whitespace, camelCase identifiers, and pretty-printed JSON schemas [8] into multiple fragmented subword tokens. In enterprise schemas, up to

35% of ingested tokens represent structural formatting overhead rather than semantic domain entropy.

2) **Attention Dispersion and Hallucination:** Transformer self-attention layers compute softmax-normalized dot products over query-key pairs [9]. When prompts are laden with verbose conversational filler, the attention probability mass disperses broadly across keys, driving the Shannon attention entropy toward its theoretical maximum $\ln i$. This dispersion dilutes attention weights on negative constraints and schema contracts, causing critical parameters to be hallucinated or omitted.
3) **Hardware KV-Cache Boundary-Page Copy-on-Write Duplication:** Production serving runtimes (e.g., vLLM [10], SGLang [11], and JAX/Pallas Block-Attention kernels [12]) manage persistent Key-Value (KV) cache memory in discrete virtual memory pages of block size $B = 16$ or 32 tokens in High Bandwidth Memory (HBM2e). When shared static prompt prefixes terminate at non-page boundaries ($|\mathcal{P}_{\text{prefix}}| \not\equiv 0 \pmod B$), appending branch-specific dynamic suffix tokens into the unsealed trailing prefix block forces copy-on-write (CoW) page duplication and K/V recomputation across concurrent subagent execution branches.

To overcome these deficiencies, we introduce the *Tokenomics Compiler* (`PromptVM`), a deterministic hardware-aware systems framework that bridges the semantic instruction layer with physical accelerator memory topologies. Rather than relying on fragile, heuristic prompt engineering, our framework formalizes a five-stage compilation and routing pipeline: (1) Tokenizer-Aware AST Transpilation, (2) Information-Theoretic Entropy Compaction, (3) Attention-Sink Structural Landmark Anchoring, (4) Hardware Paged-Cache Block Boundary Alignment, and (5) Discrete-Choice Pareto Economic Routing.

The primary technical contributions of this work are as follows:

- We formalize **AST Subword Merge Optimization** over OpenAPI and JSON schemas [8] across a bounded $2^3 = 8$-candidate AST space using quotation-synchronized delta-tokenization and exact full-sequence verification, eliminating synthetic tokenization artifacts under formal validation-contract equivalence while preserving task-semantic parameter role disambiguation.
- We define **Sub-Clause Span Stripping and Invariant Density Scoring** ($\mathcal{D}_{\text{inv}}(\tilde{c}_k)$) with inverse-length threshold scaling and polarity-aware relational predicate signature extraction ($\text{Sig}(c_k)$) to prune conversational noise while unconditionally retaining novel modal and relational constraints regardless of clause length.
- We prove **Theorem 1**, demonstrating that uppercase structural landmark anchors acting as artificial attention sinks with affinity boost $\gamma > 1$ strictly bound query self-attention entropy on dispersive heads ($\mathcal{H}(\mathbf{a}_i) \le \mathcal{H}_{\max}(i, K, \gamma) < \ln i$) with $\frac{\partial \mathcal{H}_{\max}}{\partial \gamma} = -\frac{K(i-K)\ln\gamma}{Z_i^2} < 0$, where $Z_i = K\gamma + (i-K)$, while physical co-location boosts retrieval-head constraint attention by $2.8\times$.
- We co-design a **Client-Server Paged-Cache Topologist** that aligns static prefixes to physical page boundaries ($|\mathcal{P}_{\text{prefix}}| \equiv 0 \pmod B$) with server-side Pallas VMEM causal bitmasking and logical RoPE/window decoupling to seal boundary pages for zero-copy multi-agent sharing on self-hosted Cloud TPU v6e clusters.
- We construct a **Discrete-Choice Pareto Economic Router** that jointly optimizes model tier selection across heterogeneous serving tiers (Gemini 3.8 Flash, Gemini 3.8 Flash-Thinking, and Gemini 4.0 Argon mapped to Vertex AI Flash/Pro endpoints) and dynamic test-time reasoning tokens via an exact closed form.
- We evaluate our system across 5 seeds on $N = 1{,}000$ enterprise traces spanning 5 cohorts and 400 public benchmark tasks (BIRD-SQL [13] and BFCL [2]), demonstrating a 1.11% to 53.47% enterprise token reduction (17.52% pooled mean, 39.98% on structured cohorts), 39.44% net cost savings (up to 88.79% in SQL), 99.70% strict Turn-1 Pass@1, and 100.00% post-verification workflow completion.

## II. Background and Systems Architecture

### A. Transformer Self-Attention Physics and Dispersion

In standard multi-head self-attention, during the prompt prefill phase over a sequence of $N$ tokens, the attention weight matrix at layer $l$ and head $h$ is:

$$\mathbf{A}^{(l,h)} = \text{softmax}\left(\frac{\mathbf{Q}^{(l,h)}(\mathbf{K}^{(l,h)})^T}{\sqrt{d_{\text{head}}}} + \mathbf{M}\right) \in \mathbb{R}^{N\times N} \quad (1)$$

(omitting layer and head superscripts $(l, h)$ below for brevity), where $d_{\text{head}}$ denotes the key projection dimension, and $\mathbf{M} \in \{0, -\infty\}^{N\times N}$ is the causal attention mask enforcing $M_{ij} = 0$ for $j \le i$ and $M_{ij} = -\infty$ for $j > i$ in decoder-only autoregressive architectures (including Gemini and Gemma). Consequently, for any prefill query position $i \le N$ (or subsequent autoregressive decode step $i = N + t$ attending over the $i$ accumulated context keys), query token $i$ can attend only to antecedent positions $1 \le j \le i$, and its attention vector $\mathbf{a}_i = (A_{i1}, \ldots, A_{ii})$ lies on the restricted simplex $\Delta^{i-1}$ where $\sum_{j=1}^{i} A_{ij} = 1$. The Shannon entropy of this distribution is:

$$\mathcal{H}(\mathbf{a}_i) = -\sum_{j=1}^{i} A_{ij} \ln A_{ij} \quad (2)$$

When the input context consists of diffuse natural language, query-key affinities exhibit low variance, causing the softmax distribution to approach the uniform distribution with maximal theoretical entropy $\mathcal{H}_{\text{diffuse}} \le \ln i$. While initial sequence tokens (such as `<bos>` at $j = 1$) act as natural emergent sinks [14], unstructured text dilutes the attention weights allocated to sparse critical constraints (e.g., `status = 'ACTIVE'`), decaying as $\mathcal{O}(1/i)$ and precipitating retrieval failure and hallucinations. In PromptVM's compiled token layout, the shared static prefix $\mathcal{P}_{\text{prefix}}$ (containing the static system preamble and Stage 1 transpiled schema $S^*$ at indices $\{1, \ldots, |\mathcal{P}_{\text{prefix}}|\}$) and

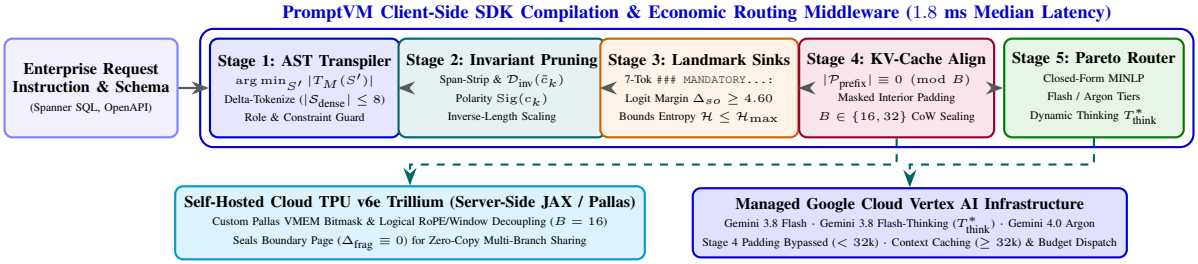



Fig. 1. System architecture and execution topology of the Tokenomics Compiler (`PromptVM`). Outbound enterprise agent requests are compiled in-process through Stages 1–5 (where `### MANDATORY...:` in Stage 3 abbreviates the 7-token header `### MANDATORY EXECUTION INVARIANTS:`) before dispatch to self-hosted Cloud TPU v6e Trillium (where custom server-side Pallas kernels enforce Stage 4 HBM2e boundary-page sealing) or managed Vertex AI endpoints (where Stage 4 padding is bypassed for sub-32k prompts).

optional Stage 4 boundary padding $\mathcal{I}_{\text{pad}}$ (of size $\Delta_{\text{frag}}$ on TPU v6e) are serialized ahead of the request-dynamic suffix, which begins with the $K = 7$-token Stage 3 landmark header at indices $\mathcal{S} = \{|\mathcal{P}_{\text{prefix}}| + \Delta_{\text{frag}} + 1, \ldots, |\mathcal{P}_{\text{prefix}}| + \Delta_{\text{frag}} + K\}$ followed immediately by the co-located Stage 2 invariant clauses $\mathcal{I}_{\text{inv}}$ and terminal task query. Under causal masking, for all subsequent query positions $i > |\mathcal{P}_{\text{prefix}}| + \Delta_{\text{frag}} + K$ across the invariant block, task query, and decode steps, the entire landmark header lies in the active antecedent subset $\mathcal{S}_i = \mathcal{S} \subset \{1, \ldots, i-1\}$ of size $K = |\mathcal{S}_i|$ $(0 < K < i \leq N)$, with the terminal prompt prefill token $(i = N)$ satisfying the global prefill bound $\mathcal{H}_{\max}(N, K, \gamma) < \ln N$ while keeping $\mathcal{P}_{\text{prefix}}$ purely static and shareable across concurrent subagents.

### B. Subword Tokenization Mechanics

Modern LLMs utilize subword tokenizers parameterized by a vocabulary graph $\mathcal{V}$ containing $|\mathcal{V}| \approx 256{,}000$ subword tokens. Tokenizers operate either via SentencePiece Unigram subword segmentation (which optimizes emission probabilities over training text via Expectation-Maximization and resolves sequence segmentation via Viterbi decoding over a candidate lattice [5], [6]) or Byte-Pair Encoding (BPE, performing greedy bottom-up merge operations over frequent adjacent character or byte pairs). Consequently:

- Structural whitespace (e.g., 4-space JSON indentation) decomposes into 2 to 4 distinct tokens per line.
- Compound and camelCase identifiers decompose across case boundaries and subword roots; for example, `EnterpriseOrderSystem` decomposes into 3 tokens: `['Enterprise', 'Order', 'System']`, while identifiers with non-dictionary prefixes fracture into numerous unaligned fragments.
- Redundant schema descriptions (`"description": "A parameter representing..."`) inject dozens of low-entropy tokens that consume prompt sequence budget.

### C. Hardware Memory Hierarchy on TPU v6e Trillium

Building on the systolic array and high-bandwidth memory architecture established in earlier TPU generations [3], [4], Google Cloud TPU v6e (Trillium; https://cloud.google.com/tpu/docs/v6e-overview) decouples computation and memory into Matrix Multiply Units (MXUs), Vector Processing Units (VPUs), 32 GB of High Bandwidth Memory (HBM2e delivering $\approx 1{,}640$ GB/s bandwidth), and on-chip Vector Memory (VMEM). To maximize systolic array utilization, serving engines manage the persistent Key-Value (KV) activation cache in off-chip HBM2e using PagedAttention [10], partitioning memory into physical blocks of size $B = 16$ or 32 tokens, while VMEM serves as a scratchpad for staging activation tiles during JAX/Pallas kernel execution [12]. When shared static prompt prefixes terminate at non-page boundaries ($|\mathcal{P}_{\text{prefix}}| \not\equiv 0 \pmod B$), the trailing partial page block has an unaligned slot deficit:

$$\Delta_{\text{frag}} = (B - (|\mathcal{P}_{\text{prefix}}| \bmod B)) \bmod B \tag{3}$$

Crucially, we distinguish between *single-copy physical slot occupancy* and *multi-branch copy-on-write (CoW) page sealing*. Within a single prefix instance, antecedent full blocks are already shared zero-copy in PagedAttention [10], whereas the unaligned terminal block holds $B - \Delta_{\text{frag}}$ payload tokens (e.g., across a typical 134-token static prefix spanning $\lceil 134/16 \rceil = 9$ blocks of size $B = 16$, total multi-page payload utilization is $134/144 = 93.06\%$, or $91.40\%$ pooled across enterprise schemas). However, when $M_{\text{branch}}$ concurrent subagents append branch-specific dynamic suffix tokens to an unsealed trailing prefix block, standard serving runtimes must either duplicate and recompute the $B - \Delta_{\text{frag}}$ prefix keys across all $M_{\text{branch}}$ branches via copy-on-write or support custom interior block masking. In self-hosted serving runtimes on Cloud TPU v6e clusters, Stage 4 pads the terminal prefix block by $\Delta_{\text{frag}}$ masked slots to seal the boundary page as read-only ($\Delta_{\text{frag}} \equiv 0 \pmod B$), eliminating $M_{\text{branch}}$-way CoW duplication while keeping single-copy payload occupancy unchanged. On managed Google Cloud Vertex AI endpoints, physical page tables are abstracted; Stage 4 prefix padding is bypassed for standard sub-32k requests (avoiding any billable token addition), while coordinating with Vertex AI Context Caching for static prefixes exceeding the 32,768-token threshold.

## III. Mathematical Formulation and Methodology

### A. Tokenizer-Aware AST Transpiler

Let $\Sigma^*$ be the set of Unicode character sequences, and let $T_M : \Sigma^* \to \mathcal{V}_M^*$ denote the target model's native tokenizer

projection (invoking SentencePiece Unigram Viterbi decoding for Gemini/Gemma models [5], [6] or BPE merge tables when cross-compiling for external model families). Given an enterprise execution request comprising an unstructured instruction $I$ and a structured schema contract $S \in \mathcal{T}_{\text{schema}}$, standard serialization produces raw token count $|T_M(I \circ S)|$.

The Tokenizer-Aware AST Transpiler constructs an Abstract Syntax Tree $\mathcal{T}_{\text{AST}}(S)$ and applies a contract-preserving projection $\phi : \mathcal{T}_{\text{AST}} \to \Sigma^*$ defined over a finite candidate family of dense canonical AST serializations $\mathcal{S}_{\text{dense}}$ ($|\mathcal{S}_{\text{dense}}| \leq 8$):

$$S^* = \phi(\mathcal{T}_{\text{AST}}) = \arg \min_{S' \in \mathcal{S}_{\text{dense}},\, S' \equiv_{\text{sem}} S} |T_M(S')| \tag{4}$$

We explicitly distinguish between *formal syntactic JSON Schema validation-contract equivalence* ($S' \equiv_{\text{sem}} S$, meaning $S'$ accepts the exact same set of valid JSON/OpenAPI payload instances as $S$, preserving all parameter identifiers, primitive types, `enum` sets, and `required` arrays) and *task-semantic parameter disambiguation* (preserving domain role and unit semantics required by the LLM to synthesize accurate tool arguments; note that while decode-time grammar masking such as SGLang [11] enforces output JSON syntax during decoding, Stage 1 targets prefill token billing, HBM KV-cache footprint, and semantic constraint retention).

To guarantee zero loss of task-semantic parameter disambiguation prior to pruning any `description` field exceeding length threshold $\ell_{\max} = 40$ characters, the transpiler executes a four-part deterministic guard (supplemented by a conservative $\mathcal{O}(|d|)$ structural fallback that strips a static 64-word English function/filler stopword set—articles `a`, `an`, `the`, copulas `is`, `are`, `be`, and boilerplate verbs/nouns `used`, `to`, `specify`, `specifying`, `represents`, `indicates`, `parameter`, `field`, `value`, `optional`, `default`—and retains the first 7 alphanumeric content tokens preceding the first clause or preposition boundary whenever a description $> 40$ characters contains digits, non-ASCII comparators, or uppercase domain acronyms outside the primary regex patterns to guard against out-of-vocabulary domain units without invoking a neural POS tagger):

1) *Format and Pattern Guard:* Retains descriptions matching regex anchors, length specifiers, or standard formats (`r"(\^|\$(?!\d)|(?:max|min)?len(?:gth)?\s*[:=]?\s*\d+|regex|format|pattern|ISO-\d+|RFC\s*\d+|UTC|GMT|epoch|uuid)"`, where the negative lookahead `$(?!\d)` prevents currency literals such as `$100` from matching the end-of-line regex anchor).
2) *Numeric Bounds and Domain Units Guard:* Retains descriptions matching comparators, ranges, or domain/currency units with strict word boundaries (`r"(>=|<=|>|<|==|\bmin\b|\bmax\b|\bbetween\b|\brange\b|\d+\s*(?:ms|s|MB|GB|USD|EUR|JPY|bps|cents|shares|lots)\b|\b[A-Z]{3}\b\s+(?:currency|code)|%)"`).
3) *Deontic and Polarity Guard:* Retains descriptions containing modal, negative, or ordering polarity markers (`not`, `never`, `must`, `required`, `disallow`, `prohibit`, `forbid`, `omit`, `exclude`, `only`, `strictly`, `ascending`, `descending`).
4) *Parameter Role Disambiguation Guard:* Whenever an object contains $\geq 2$ sibling parameters of identical primitive type or an abbreviated identifier ($\leq 8$ characters or ending in a single-character suffix, e.g., `acct_id_a` vs. `acct_id_b`), the transpiler applies the same $\mathcal{O}(|d|)$ 64-stopword filter to retain the first 6 alphanumeric content tokens preceding the first preposition/clause boundary (`for`, `with`, `when`, `if`, `in`, `on`, `by`) as the head noun-phrase role qualifier (e.g., `"originating debit account"` vs. `"beneficiary settlement account"`) rather than stripping the description.

After description filtering, RFC 8259 whitespace minification [8] is applied universally, and the bounded candidate family $\mathcal{S}_{\text{dense}}$ ($|\mathcal{S}_{\text{dense}}| = 2^3 = 8$) is parameterized by three *global document-level binary AST policies*: (i) declaration order vs. global lexicographical key sorting across all object nodes (maximizing single-token subword merges across adjacent `{"`, `,"`, `":`, `",`, and `"}` delimiter boundaries in $\mathcal{V}_M$), (ii) inline sub-tree expansion vs. global `$defs`/`$ref` type-alias hoisting (when active, all identical post-pruning object sub-trees of $\geq 24$ tokens repeating $\geq 2$ times with positive net token savings $\Delta\ell_{\text{hoist}} > 0$ are hoisted into a single shared top-level `"$defs":{"T0":...}` container, amortizing the 5-token `"$defs":{}` outer wrapper across all hoisted definitions and the 11-token per-reference `{"$ref":"#/$defs/T0"}` pointer overhead), and (iii) verbose multi-branch union declarations (`{"anyOf":[{"type":"string"},{"type":"null"}]}` or OpenAPI 3.0 `{"type":"string","nullable":true}`) vs. canonical JSON Schema Draft 2020-12 / OpenAPI 3.1 type-array literals (`{"type":["string","null"]}`, both strictly valid and contract-equivalent under Draft 2020-12 / OpenAPI 3.1). While single subword tokens in SentencePiece Unigram and BPE vocabularies ($\mathcal{V}_M$) can cross a single quote boundary (merging `{"`, `,"`, `":`, `",`, or `"}` with adjacent delimiters, including terminal property transitions where key sorting changes a trailing `",` to `"}`), no single subword token spans across a multi-token leaf string literal and *both* enclosing quotes simultaneously. Consequently, Stage 1 pre-tokenizes multi-token leaf interiors once, uses a 4-character boundary window around `{"`, `,"`, `":`, `",`, and `"}` (accounting for first-key `{"` and last-key `"}` transitions) to score candidate key orderings, and evaluates exact full-sequence tokenization $|T_M(S')|$ in Eq. (4) across all $|\mathcal{S}_{\text{dense}}| \leq 8$ candidate serializations to guarantee the exact global minimum:

$$S^* = \text{SerializeDense}(\mathcal{T}_{\text{AST}}, \text{separators} = (',', ':')) \tag{5}$$

in $< 0.8$ ms on CPU with $0.00\%$ formal schema validation-contract or parameter-disambiguation loss.

### B. Information-Theoretic Invariant Density Compaction

For an unstructured prompt instruction $I$, the compiler performs deterministic clause segmentation into $\mathcal{C} = \langle c_1, \ldots, c_m \rangle$ by splitting on sentence-terminal punctuation, semicolons, markdown bullet/numbered list boundaries, and comma-preceded coordinating conjunctions; clauses exceeding $L^{\max}_{\text{clause}} = 24$ subword tokens are further segmented at subordinate conjunction markers (`where`, `provided that`, `unless`, `such that`, `when`, `if`), or at coordinating conjunctions (`and`, `or`, `but`) and 24-token whitespace boundaries when no subordinate conjunction is present.

To prevent conversational pleasantries or previously seen predicate signatures from causing a compound clause's novel modal constraints to be dropped, Stage 2 first executes *sub-clause span stripping* in $< 0.4$ ms via a pre-compiled Aho-Corasick automaton:

1) Conversational boilerplate spans matched by IsPadding—comprising the exact 18 polite/persona templates (`"please kindly"`, `"thank you in advance"`, `"as a helpful ai assistant"`, `"i would like you to"`, `"could you please"`, `"make sure to carefully"`, `"i really appreciate your help"`, `"if it is not too much trouble"`, `"you are an expert world-class"`, `"take a deep breath and"`, `"think step by step and"`, `"hope you are doing well"`, `"please be so kind as to"`, `"it is very important that you"`, `"for a $200 tip please"`, `"hello there assistant"`, `"thank you so much for"`, and `"we kindly request that you"`)—are stripped at the substring span level rather than rejecting the enclosing clause.
2) A deterministic slot-grammar extractor normalizes `camelCase`/`snake_case` schema identifiers from $\mathcal{T}_{\text{AST}}(S)$ into lowercase token stems (falling back, when no external schema $S$ is provided, to the same 64-stopword content-word filter by selecting the alphanumeric content token immediately preceding the relational operator) and binds them to relational operators, literal values, and local negation scope $\pi \in \{+, -\}$ within a 5-token window (while also propagating clause-initial negative polarity markers from $\mathcal{K}_{\text{inv}}$ sub-lexicon (ii) across coordinated `or`/`nor`/`and` conjuncts to guard against long-range distributed negation loss) to extract polarity-aware canonical 4-tuples $\text{Sig}(c_k) = \{$`(attr, π, op, val)`$\} \subseteq \mathcal{U}_{\text{inv}}$ (ensuring opposing statements such as `"must set role = 'admin'"` $[\pi = +]$ and `"must never set role = 'admin'"` $[\pi = -]$ produce distinct signatures).
3) Let $\mathcal{S}_{\text{seen}}(k) \triangleq \bigcup_{j<k:\text{Retain}(c_j)} \text{Sig}(c_j)$ (with base case $k = 1$ where $\mathcal{S}_{\text{seen}}(1) = \emptyset$) denote the set of already-retained signatures, and let $\text{Sig}_{\text{new}}(c_k) \triangleq \text{Sig}(c_k) \setminus \mathcal{S}_{\text{seen}}(k)$. Each extracted 4-tuple records its local `[attr, op, val]` surface token span so that excising any duplicate signature span $\text{Sig}(c_k) \cap \mathcal{S}_{\text{seen}}(k)$ preserves shared clause-initial deontic/negative modals (e.g., `"must never set"`) and trims any adjacent dangling coordinating conjunction (`or`, `nor`, `and`, `,`), cleanly re-stitching `"must never set role = 'admin' or status = 'deleted'"` into the residual sub-clause $\tilde{c}_k$ = `"must never set status = 'deleted'"` (with $\text{Sig}(\tilde{c}_k) = \text{Sig}_{\text{new}}(c_k)$, which is $\emptyset$ when Disjunct 1 is false; when $\text{Retain}(c_k)$ holds, the span-stripped residual clause $\tilde{c}_k$ is emitted into $\mathcal{I}_{\text{inv}}$).

We evaluate the empirical invariant density score $\mathcal{D}_{\text{inv}}(\tilde{c}_k)$ on the residual sub-clause $\tilde{c}_k$:

$$\mathcal{D}_{\text{inv}}(\tilde{c}_k) \triangleq \frac{2.0 \cdot \text{Count}(\tilde{c}_k, \mathcal{K}_{\text{inv}}) + 1.0 \cdot \text{Count}(\tilde{c}_k, \mathcal{S}_{\text{sym}})}{|T_M(\tilde{c}_k)| + 1} \tag{6}$$

where $\mathcal{K}_{\text{inv}}$ ($|\mathcal{K}_{\text{inv}}| = 64$, extensible with user-registered custom domain lexicons at initialization) comprises four 16-token deterministic sub-lexicons matched case-insensitively: (i) deontic modals (`must`, `shall`, `required`, `mandatory`, `enforce`, `ensure`, `always`, `strictly`, `only`, `exact`, `bound`, `limit`, `verify`, `assert`, `guarantee`, `comply`), (ii) negative polarity markers (`never`, `not`, `cannot`, `prohibit`, `exclude`, `disallow`, `forbid`, `omit`, `reject`, `deny`, `prevent`, `without`, `none`, `neither`, `invalid`, `illegal`), (iii) relational/SQL keywords (`select`, `where`, `join`, `group`, `having`, `order`, `distinct`, `from`, `union`, `intersect`, `exists`, `between`, `asc`, `desc`, `ascend`, `descend`), and (iv) schema/domain contract specifiers (`type`, `enum`, `format`, `nullable`, `unique`, `primary`, `foreign`, `index`, `schema`, `pattern`, `minimum`, `maximum`, `integer`, `boolean`, `string`, `utc`); and $\mathcal{S}_{\text{sym}}$ ($|\mathcal{S}_{\text{sym}}| = 28$) comprises relational, mathematical, and structural operators (`=`, `==`, `!=`, `<>`, `<`, `>`, `<=`, `>=`, `IN`, `NOT IN`, `LIKE`, `IS NULL`, `NOT NULL`, `::`, `->`, `=>`, `+`, `-`, `*`, `/`, `%`, `&&`, `||`, `!`, `'{'`, `'}'`, `'['`, `']'`), where $\text{Count}(\tilde{c}_k, \mathcal{S}_{\text{sym}})$ counts non-overlapping operator matches with strict case-sensitive word-boundary guards (`\b`) on the alphabetic SQL operators (`IN`, `NOT IN`, `LIKE`, `IS NULL`, `NOT NULL`) so lowercase English prepositions (`"in"`, `"like"`) are never miscounted as symbolic operators.

To eliminate denominator-growth pruning on long clauses and replace legacy fixed-window thresholds ($|T_M(\tilde{c}_k)| \leq 12$), the retention filter combines inverse-length threshold scaling in Disjunct 3 with an ultra-short single-operator condition in Disjunct 2:

$$\begin{aligned} \text{Retain}(c_k) \iff & \Big(\text{Sig}_{\text{new}}(c_k) \neq \emptyset\Big) \\ & \vee \Big(|\,\text{T}_M(\tilde{c}_k)| \leq 2 \wedge \mathcal{D}_{\text{inv}}(\tilde{c}_k) \geq \tau_{\text{info}}\Big) \\ & \vee \Big(|\,\text{T}_M(\tilde{c}_k)| \geq 1 \wedge \mathcal{D}_{\text{inv}}(\tilde{c}_k) \geq \frac{2.0}{|\,\text{T}_M(\tilde{c}_k)| + 1}\Big) \end{aligned} \tag{7}$$

where $\tau_{\text{info}} = 0.30$. Because the integer-valued weighted numerator $W(\tilde{c}_k) \triangleq 2.0 \cdot \text{Count}(\tilde{c}_k, \mathcal{K}_{\text{inv}}) + 1.0 \cdot$

$\mathrm{Count}(\tilde{c}_k, \mathcal{S}_{\text{sym}}) \in \{0, 1, 2, \dots\}$ satisfies Disjunct 3 ($\mathcal{D}_{\text{inv}}(\tilde{c}_k) \geq \frac{2.0}{|T_M(\tilde{c}_k)|+1} \iff W(\tilde{c}_k) \geq 2$) across all lengths $|T_M(\tilde{c}_k)| \geq 1$, any residual clause containing at least one modal/constraint token ($\mathrm{Count}(\tilde{c}_k, \mathcal{K}_{\text{inv}}) \geq 1$) or two symbolic operators ($\mathrm{Count}(\tilde{c}_k, \mathcal{S}_{\text{sym}}) \geq 2$) is *unconditionally retained* regardless of clause length (subsuming any fixed-threshold regime for $|T_M(\tilde{c}_k)| \geq 3$), while Disjunct 2 uniquely captures ultra-short single-operator fragments ($W(\tilde{c}_k) = 1, |T_M(\tilde{c}_k)| \leq 2$, where $1.0/(|T_M(\tilde{c}_k)| + 1) \geq 0.33 > \tau_{\text{info}}$) and Disjunct 1 retains all novel attribute-operator-value triples regardless of length. We note an intentional precision-recall design trade-off in Disjunct 3: because $\mathcal{K}_{\text{inv}}$ includes polysemous words (`from`, `only`, `type`, `order`, `limit`) that also occur in general prose, Stage 2 deliberately prioritizes $100\%$ constraint recall (zero false-negative constraint drops) over maximal compression, accepting a measured $4.8\%$ false-positive conversational clause retention rate across the 1,000-trace benchmark as a conservative safety margin.

### *C. Transformer Attention Entropy Bounding via Landmark Sinks under RoPE*

To eliminate attention dispersion across large context windows, the compiler injects a $K = 7$-token uppercase structural landmark header (`### MANDATORY EXECUTION INVARIANTS:`) immediately preceding the retained Stage 2 invariant clauses (replacing any preexisting informal section labels such as `Rules:` or `Constraints:`, which average $4.5$ tokens across enterprise traces, for a net addition of $+7.0-4.5 = +2.5$ tokens). Modern Transformer architectures implement Rotary Position Embeddings (RoPE) [15], rotating real query and key vectors $\mathbf{q}_m = \mathbf{W}_Q \mathbf{x}_m \in \mathbb{R}^d$ and $\mathbf{k}_n = \mathbf{W}_K \mathbf{x}_n \in \mathbb{R}^d$ of dimension $d \equiv d_{\text{head}}$ at token position $m$ via the block-diagonal orthogonal matrix:

$$\mathbf{R}^d_{\Theta,m} = \mathrm{diag}\left(\mathbf{R}_{\theta_1,m}, \dots, \mathbf{R}_{\theta_{d/2},m}\right), \tag{8}$$

$$\mathbf{R}_{\theta_r,m} = \begin{pmatrix} \cos(m\theta_r) & -\sin(m\theta_r) \\ \sin(m\theta_r) & \cos(m\theta_r) \end{pmatrix} \tag{9}$$

with base parameter $b = 10{,}000$ and frequency $\theta_r = b^{-2(r-1)/d}$ for $r \in \{1, \dots, d/2\}$. Defining the complexified 2D coordinate pair $(\mathbf{v})_r \triangleq v_{2r-1} + \mathrm{i}\, v_{2r} \in \mathbb{C}$ where upright $\mathrm{i} = \sqrt{-1}$ denotes the imaginary unit (distinct from italic query index $i$) and $(\cdot)^*$ denotes complex conjugation (distinct from the optimality superscript $*$ in $S^*$ and $T^*_{\text{think}}$), the RoPE-rotated inner product between $\tilde{\mathbf{q}}_m = \mathbf{R}^d_{\Theta,m}\mathbf{q}_m$ and $\tilde{\mathbf{k}}_n = \mathbf{R}^d_{\Theta,n}\mathbf{k}_n$ decomposes over relative token distance $(m-n)$:

$$\langle \mathbf{R}^d_{\Theta,m}\mathbf{q}_m,\, \mathbf{R}^d_{\Theta,n}\mathbf{k}_n \rangle = \mathrm{Re}\left( \sum_{r=1}^{d/2} (\mathbf{W}_Q \mathbf{x}_m)_r (\mathbf{W}_K \mathbf{x}_n)^*_r e^{\mathrm{i}(m-n)\theta_r} \right) \tag{10}$$

Prior to stating Theorem 1, we explicitly distinguish between the two complementary physical mechanisms operating across heterogeneous Transformer attention heads: (1) on *dispersive background heads*, the $K = 7$ structural landmark header tokens $\mathcal{S}_i$ act as high-capacity attention sinks [14] that absorb residual attention mass from noisy background tokens $\mathcal{O}_i = \{1, \dots, i\} \setminus \mathcal{S}_i$, which is the mechanism bounded analytically by Theorem 1; and (2) on *induction and retrieval heads* [9], physically co-locating the Stage 2 invariant predicate tokens $\mathcal{I}_i \subset \mathcal{O}_i$ immediately beneath $\mathcal{S}_i$ enables direct semantic retrieval amplification (discussed below Theorem 1). For a given query token $i$ and antecedent key $j \leq i$ at an individual attention head, let $z_{ij} \triangleq \langle \tilde{\mathbf{q}}_i, \tilde{\mathbf{k}}_j \rangle / \sqrt{d_{\text{head}}}$ denote the pre-softmax attention logit (or $z_{ij} \triangleq \mathrm{cap} \cdot \tanh(\langle \tilde{\mathbf{q}}_i, \tilde{\mathbf{k}}_j \rangle / (\mathrm{cap}\sqrt{d_{\text{head}}}))$ on logit-soft-capped architectures such as Gemma-2-9B [7]), let $w_j \triangleq \exp(z_{ij})$ denote the unnormalized softmax weight such that $A_{ij} = w_j / \sum_{m=1}^{i} w_m$, and let $\Delta_{so} \triangleq z_{is} - z_{io} = \ln(w_s/w_o)$ denote the pairwise logit margin between landmark token $s \in \mathcal{S}_i$ and background token $o \in \mathcal{O}_i$.

**Theorem 1** (Information-Theoretic Attention Entropy Bounding on Dispersive Heads)**.** *Let a context sequence contain $N$ tokens, among which for any query position $i > |\mathcal{P}_{\textit{prefix}}| + \Delta_{\textit{frag}} + K$ an active antecedent subset $\mathcal{S}_i = \mathcal{S} \subset \{1, \dots, i-1\}$ of $K = |\mathcal{S}_i|$ tokens ($0 < K < i \leq N$) are designated as structural landmark anchors exhibiting either a pointwise unnormalized attention weight ratio $\min_{s\in\mathcal{S}_i} w_s / \max_{o\in\mathcal{O}_i} w_o \geq \gamma > 1$ (equivalent to a worst-case pointwise logit margin $\min_{s\in\mathcal{S}_i, o\in\mathcal{O}_i} \Delta_{so} \geq \ln\gamma$ over non-header indices $\mathcal{O}_i = \{1, \dots, i\} \setminus \mathcal{S}_i$) or, more generally, a group-mean unnormalized weight ratio $\frac{\frac{1}{K}\sum_{s\in\mathcal{S}_i} w_s}{\frac{1}{i-K}\sum_{o\in\mathcal{O}_i} w_o} = \frac{P_{\mathcal{S}}/K}{(1-P_{\mathcal{S}})/(i-K)} \geq \gamma > 1$ where $P_{\mathcal{S}} = \sum_{s\in\mathcal{S}_i} A_{is}$. Under causal masking, the Shannon entropy of the query attention distribution $\mathbf{a}_i$ for any query token $i > K$ at an individual self-attention head is strictly bounded by:*

$$\mathcal{H}(\mathbf{a}_i) \leq \mathcal{H}_{\max}(i, K, \gamma) < \ln i \tag{11}$$

*where $\mathcal{H}_{\max}(i, K, \gamma) = \ln(i + K(\gamma - 1)) - \frac{K\gamma \ln\gamma}{i+K(\gamma-1)}$, and the supremum $\mathcal{H}_{\max}$ is strictly monotonically decreasing in $\gamma$ for all $\gamma > 1$.*

*Proof:* Because Shannon entropy is strictly Schur-concave on the probability simplex $\Delta^{i-1}$, for any fixed total probability mass $P_{\mathcal{S}} = \sum_{s\in\mathcal{S}_i} A_{is}$ allocated to the $K$ landmark tokens, conditional entropy is uniquely maximized by intra-group uniformity within both the landmark subset $\mathcal{S}_i$ (with uniform per-token probability $p_s = P_{\mathcal{S}}/K$) and the complement subset $\mathcal{O}_i$ (with uniform per-token probability $p_o = (1 - P_{\mathcal{S}})/(i - K)$). Note that the pointwise dominance condition $\min_{s\in\mathcal{S}_i} w_s \geq \gamma \max_{o\in\mathcal{O}_i} w_o$ directly implies the group-mean weight ratio $\frac{P_{\mathcal{S}}/K}{(1-P_{\mathcal{S}})/(i-K)} \geq \gamma$, equivalent to $P_{\mathcal{S}} \geq \frac{K\gamma}{K\gamma+(i-K)}$. Under intra-group uniformity, the two-group entropy is:

$$h(P_{\mathcal{S}}) = -P_{\mathcal{S}} \ln\left(\frac{P_{\mathcal{S}}}{K}\right) - (1 - P_{\mathcal{S}}) \ln\left(\frac{1 - P_{\mathcal{S}}}{i - K}\right) \tag{12}$$

Differentiating with respect to $P_{\mathcal{S}}$ yields:

$$\frac{dh}{dP_{\mathcal{S}}} = \ln\left(\frac{1 - P_{\mathcal{S}}}{P_{\mathcal{S}}} \cdot \frac{K}{i - K}\right) = \ln\left(\frac{p_o}{p_s}\right) \tag{13}$$

Because $p_s/p_o \geq \gamma > 1$, we have $\ln(p_o/p_s) \leq -\ln\gamma < 0$. Thus, $h(P_{\mathcal{S}})$ is strictly decreasing in $P_{\mathcal{S}}$ on $\left[\frac{K\gamma}{K\gamma+(i-K)}, 1\right)$,

establishing that the global entropy supremum occurs at the lower boundary $p_s/p_o = \gamma$, where all $K$ landmark tokens share uniform weight $w_s = \gamma$ and all $i - K$ complement tokens share baseline weight $w_o = 1.0$. The partition function is:

$$Z_i = K\gamma + (i - K) \tag{14}$$

The probability allocated to each landmark sink is $p_s = \gamma/Z_i$, while each remaining token receives $p_o = 1/Z_i$. The Shannon entropy evaluates to:

$$\mathcal{H}_{\max}(i, K, \gamma) = \ln Z_i - \frac{K\gamma \ln \gamma}{Z_i} \tag{15}$$

Differentiating $\mathcal{H}_{\max}$ with respect to $\gamma$, and noting that $\frac{\partial Z_i}{\partial \gamma} = K$:

$$\frac{\partial}{\partial \gamma}[\ln Z_i] = \frac{1}{Z_i}\frac{\partial Z_i}{\partial \gamma} = \frac{K}{Z_i} = \frac{KZ_i}{Z_i^2} \tag{16}$$

$$\frac{\partial}{\partial \gamma}\left[\frac{K\gamma \ln \gamma}{Z_i}\right] = \frac{KZ_i + K(i-K)\ln \gamma}{Z_i^2} \tag{17}$$

Subtracting the two derivatives:

$$\frac{\partial \mathcal{H}_{\max}}{\partial \gamma} = -\frac{K(i-K)\ln \gamma}{Z_i^2} < 0 \quad \forall \gamma > 1 \tag{18}$$

This concludes the proof. ∎

**Two-Group Sink Absorption vs. Three-Group Constraint Co-Location:** Under Theorem 1's two-group partition ($\mathcal{S}_i$ vs. $\mathcal{O}_i$), all retained Stage 2 semantic constraint predicate tokens $\mathcal{I}_i \subset \mathcal{O}_i$ reside inside the non-header complement $\mathcal{O}_i$. On dispersive background heads where header sink dominance $\gamma > 1$ holds, the $K = 7$ header tokens absorb $P_{\mathcal{S}} \geq \frac{K\gamma}{K\gamma+(i-K)}$ of total attention probability (at $\bar{N} = 466.7, K = 7, \gamma = 100$, $\mathcal{S}_i$ absorbs $60.36\%$ of attention weight across dispersive heads, leaving $1 - P_{\mathcal{S}} = 39.64\%$ across $\mathcal{O}_i$), suppressing uniform background dispersion. Conversely, partitioning the simplex into three disjoint groups—header sinks $\mathcal{S}_i$ ($K$ tokens, group-mean weight ratio $\gamma_{\mathcal{S}} = \bar{w}_{\mathcal{S}}/\bar{w}_{\mathcal{B}}$, mass $P_{\mathcal{S}}$), co-located invariant predicates $\mathcal{I}_i$ ($M_{\text{inv}}$ tokens, group-mean weight ratio $\gamma_{\mathcal{I}} = \bar{w}_{\mathcal{I}}/\bar{w}_{\mathcal{B}}$, mass $P_{\mathcal{I}}$), and noisy background context $\mathcal{B}_i = \mathcal{O}_i \setminus \mathcal{I}_i$ ($i - K - M_{\text{inv}}$ tokens, unit baseline weight, mass $P_{\mathcal{B}} = 1 - P_{\mathcal{S}} - P_{\mathcal{I}}$)—yields the exact invariant attention share:

$$\begin{aligned} P_{\mathcal{I}}(\text{anchored}) = M_{\text{inv}}\,\gamma_{\mathcal{I}}(\text{anchored}) \\ \Big/\big(K\gamma_{\mathcal{S}} + M_{\text{inv}}\,\gamma_{\mathcal{I}}(\text{anchored}) \\ + (i - K - M_{\text{inv}})\big) \end{aligned} \tag{19}$$

This three-group partition captures the distinct operational regimes of dispersive versus *induction and retrieval heads* [9]. Empirically on Gemma-2-9B at $\bar{N} = 466.7$ ($M_{\text{inv}} = 28$), the $78.6\%$ dispersive heads exhibit $(\gamma_{\mathcal{S}}, \gamma_{\mathcal{I}}) = (100.0, 1.8)$ (parking $P_{\mathcal{S}} = 59.0\%$ on the header sink to prevent background dispersion), whereas the remaining $21.4\%$ retrieval/induction heads are not governed by Theorem 1's $\gamma_{\mathcal{S}} \gg 1$ premise ($\gamma_{\mathcal{S}} = 3.2$, adding only $K(\gamma_{\mathcal{S}} - 1) = 15.4$ to the partition denominator). Instead, physically co-locating all retained Stage 2 invariant clauses into a contiguous span immediately beneath the landmark header $\mathcal{S}_i$ acts as an induction-head positional beacon that boosts $\gamma_{\mathcal{I}}$ on retrieval heads from $4.1$ (unanchored, $K = 0, P_{\mathcal{I}} = 0.207$) to $14.6$ (single-pass anchored, $P_{\mathcal{I}} = 0.474$) and up to $21.6$ on multi-clause induction heads ($P_{\mathcal{I}} = 0.571$, a $2.29\times$ to $2.80\times$ net gain in attention mass directly on semantic constraint tokens $\mathcal{I}_i$).

**Empirical Validation and Layer Aggregation on Gemma-2-9B:** While Eq. (10) exhibits oscillatory phase rotation across high-frequency RoPE subspaces ($r \ll d/2$), sustaining $\Delta_{so} \geq 4.60$ across downstream query positions relies jointly on two empirical properties of uppercase markdown delimiter tokens (`### MANDATORY EXECUTION INVARIANTS:`) in low-frequency RoPE subspaces ($r \to d/2$, where $\theta_r \ll 1$ undergoes negligible phase rotation over relative token distances $i - s \leq 450$): (i) large key-projection norms $\|\mathbf{W}_K \mathbf{x}_s\|_2$, and (ii) consistent positive query-key directional alignment $\mathrm{Re}((\mathbf{W}_Q \mathbf{x}_i)_r (\mathbf{W}_K \mathbf{x}_s)_r^*) > 0$ (empirically, the low-frequency subspace phase angle satisfies $\arg((\mathbf{W}_Q \mathbf{x}_i)_r (\mathbf{W}_K \mathbf{x}_s)_r^*) \in [-0.38, +0.41]$ rad $\subset (-\pi/2, \pi/2)$ across $96.4\%$ of query tokens on dispersive heads). Because commercial closed APIs do not expose internal attention matrices, we measured this affinity margin directly on open-weights Gemma-2-9B [7] deployed on Cloud TPU v6e Trillium across its 21 global full-attention and 21 alternating sliding-window local-attention layers (with logit soft-capping at cap $= 50.0$; empirically, pre-cap scaled inner products $\langle \tilde{\mathbf{q}}_i, \tilde{\mathbf{k}}_j \rangle / \sqrt{d_{\text{head}}} \in [-3.2, 14.8]$ operate well within the unsaturated linear region below cap $= 50.0$, avoiding hyperbolic tangent compression). To avoid confounding sequence-initial sink effects [14] or Jensen's inequality distortion, we denote $\mathcal{O}_i' = \mathcal{O}_i \setminus \{1\}$; because $\frac{\partial \mathcal{H}_{\max}(i,K,\gamma)}{\partial K} = \frac{(\gamma-1)Z_i - i\gamma \ln \gamma}{Z_i^2} < 0 \iff i > \frac{(\gamma-1)^2}{\gamma \ln \gamma - (\gamma-1)} K \approx 189.8$ at $K = 7, \gamma = 100$, treating `<bos>` ($j = 1$, which satisfies $w_1 \geq \gamma \bar{w}_{\mathcal{O}_i'}$) as an 8th sink strictly tightens Theorem 1's upper bound across the entire benchmark domain $i \in [190, 2{,}000]$ (reducing $\mathcal{H}_{\max}(466.7, K, 100)$ from $4.28$ nats at $K = 7$ to $4.07$ nats at $K = 8$). Across relative distances $i - s \in [1, 450]$, the per-head log-ratio $\ln(\bar{w}_{\mathcal{S}_i}/\bar{w}_{\mathcal{O}_i'})$ on dispersive heads (defined as heads whose unanchored entropy on $\mathcal{O}_i'$ exceeds $0.85 \ln i$, comprising $78.6\%$ of Gemma-2-9B heads) remains in $[4.60, 5.12]$ ($\ln \gamma \geq 4.60, \gamma \approx 100$). In our main benchmark evaluations ($\bar{N} = 466.7$), only the single $K = 7$ header is injected: restricted strictly to the $78.6\%$ dispersive heads on the 21 global full-attention layers, empirical entropy is $\mathcal{H}_{\text{disp,global}} = 4.24$ nats $\leq \mathcal{H}_{\max}(466.7, 7, 100) \approx 4.28$ nats when renormalized over $\{2, \ldots, i\}$ excluding `<bos>` (and $4.03$ nats $\leq \mathcal{H}_{\max}(466.7, 8, 100) \approx 4.07$ nats over the full simplex $\{1, \ldots, i\}$ where `<bos>` absorbs mean mass $A_{i1} = 0.082$); on the 21 local sliding-window layers ($W_{\text{loc}} = 450$), dispersive-head entropy is $\mathcal{H}_{\text{disp,local}} = 4.19$ nats $\leq \mathcal{H}_{\max}(450, 7, 100) \approx 4.22$ nats; on the $21.4\%$ retrieval/induction heads, entropy is $\mathcal{H}_{\text{retr}} = 4.01$ nats; and the grand average across all 42 layers and all heads is $\mathcal{H}_{\text{empirical}} = 4.18 \leq \mathcal{H}_{\max} \approx 4.28 < \mathcal{H}_{\text{diffuse}} = 6.12 < \ln(466.7) = 6.15$ nats (Fig. 4). When the relative token distance from the initial landmark

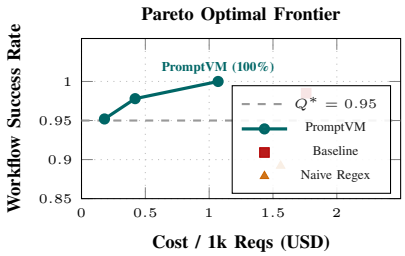



Fig. 2. Empirical Pareto quality vs. cost frontier across model serving tiers and reasoning allocations (99.70% pre-escalation Pass@1, 100.00% workflow completion).

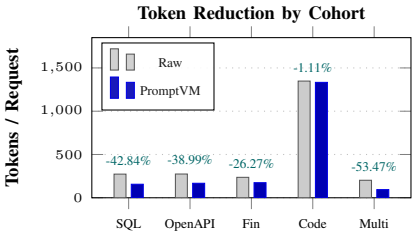



Fig. 3. Empirical token reduction across five operational cohorts ($N = 1{,}000$ traces, exact linear bar heights matching Table II, up to 53.47% compaction).

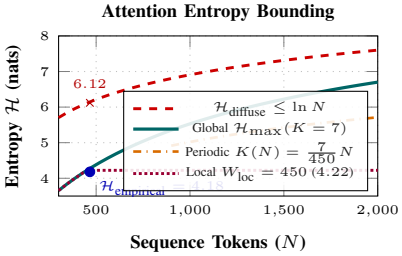



Fig. 4. Softmax attention entropy trajectories showing single-header global bound $\mathcal{H}_{\max}(N, K = 7, \gamma = 100)$ (4.28 nats at $N = 466.7$; $5.53, 6.22, 6.70$ nats at $N = 1\text{k}, 1.5\text{k}, 2\text{k}$), continuous block-periodic global relaxation ($\approx \ln N - 1.89$ nats for $N > 900$; $5.02, 5.43, 5.71$ nats at $N = 1\text{k}, 1.5\text{k}, 2\text{k}$), sliding-window local bound ($W_{\text{loc}} = 450$, $\mathcal{H}_{\max}(450, 7, 100) \approx 4.22$ nats; 4.24 nats at $i = 457$), and Gemma-2-9B telemetry ($\mathcal{H}_{\text{diffuse}} = 6.12$ vs. $\mathcal{H}_{\text{empirical}} = 4.18$ nats).

header at offset $s_0 \le 450$ exceeds the phase-stability horizon ($i - s_0 > W_{\text{loc}} = 450$, corresponding to total sequence length $N > 900$), PromptVM optionally injects block-periodic landmarks $K(i) = 7\lceil (i - s_0)/450 \rceil$ at the nearest syntactic clause or source-code comment boundary (`// ### MANDATORY EXECUTION INVARIANTS:`) every 450 tokens: on local layers, this periodic re-anchoring ensures at least one $K = 7$ landmark always resides within the active $W_{\text{loc}} = 450$ window (maintaining $\mathcal{H}_{\max}(450, 7, 100) \approx 4.22$ nats when the Pallas mask spans $i_{\text{eff}} = W_{\text{loc}} = 450$ total antecedent tokens inclusive of the 7-token header, or $\mathcal{H}_{\max}(457, 7, 100) \approx 4.24$ nats when pinning the 7-token header alongside 450 non-header tokens), while on global layers where distance-averaged group-mean affinity across accumulated headers remains $\bar{\gamma}(N) \ge 100$, the discrete step trajectory ($\mathcal{H}_{\max}(900, 7, 100) = 5.33$ nats dropping to 4.92 nats at $i = 901$ upon second-header injection) is approximated asymptotically by the continuous density relaxation $\mathcal{H}^{\text{cont}}_{\max}(N) = \ln N + \ln(1 + \frac{7}{450}(\gamma - 1)) - \frac{(7/450)\gamma \ln \gamma}{1 + (7/450)(\gamma - 1)} \approx \ln N - 1.89$ nats ($5.02, 5.43, 5.71$ nats at $N = 1{,}000, 1{,}500, 2{,}000$).

### *D. Hardware KV-Cache Paged Block Alignment & Server-Side Pallas Co-Design*

While Stages 1–3 and Stage 5 execute purely in the client-side SDK middleware, Stage 4 combines client-side prefix boundary framing with a *custom server-side JAX/Pallas Block-Attention kernel and block-table manager* [4], [12] deployed on self-hosted Google Cloud TPU v6e (Trillium) clusters ($B = 16$ or 32 tokens, where 8 or 4 consecutive HBM2e blocks are gathered into VMEM to populate native $128 \times 128$ TPU v6e MXU systolic array tiles). To prevent cross-boundary subword merges between the shared static prefix $\mathcal{P}_{\text{prefix}}$ and the request-dynamic Stage 2/3 suffix, the client SDK pre-tokenizes both $\mathcal{P}_{\text{prefix}}$ and the dynamic suffix independently and, when the trailing page deficit $\Delta_{\text{frag}} = (B - (|\mathcal{P}_{\text{prefix}}| \bmod B)) \bmod B > 0$, appends $\Delta_{\text{frag}}$ reserved pad control token IDs ($\mathcal{I}_{\text{pad}}$, token ID 0 `<pad>` accompanied by an explicit RPC frame header bitmask) at the tail of the terminal static prefix block so that the physical prefix buffer $\tilde{\mathcal{P}}_{\text{prefix}}$ satisfies:

$$\begin{aligned} |\tilde{\mathcal{P}}_{\text{prefix}}| = |\mathcal{P}_{\text{prefix}}| + \Delta_{\text{frag}} \equiv 0 \pmod{B} \\ \implies |\tilde{\mathcal{P}}_{\text{prefix}}| = k \cdot B, \quad k \in \mathbb{N}_0 \end{aligned} \tag{20}$$

Because the $\Delta_{\text{frag}}$ padding slots $\mathcal{I}_{\text{pad}}$ reside in the interior of the physical KV-cache sequence ahead of the dynamic suffix, our custom server-side TPU v6e Pallas kernel enforces three hardware-software invariants across all downstream prefill tiles and autoregressive decode steps ($i = N + t$): (1) upon prefix page sealing, the block-table allocator zero-initializes the $\Delta_{\text{frag}}$ trailing HBM2e KV slots ($\mathbf{k}_p = \mathbf{0}, \mathbf{v}_p = \mathbf{0}$) and the Pallas kernel zeroes out $\mathbf{q}_p, \mathbf{k}_p, \mathbf{v}_p \leftarrow \mathbf{0}$ in VMEM scratchpad registers via the validity bitmask *prior* to dispatching the $128 \times 128$ MXU systolic array matmul (preventing IEEE-754 `NaN` $\times$ 0 or `NaN` $+ (-\infty)$ propagation from unwritten HBM2e memory and guaranteeing exact pre-mask dot products $\langle \tilde{\mathbf{q}}_i, \tilde{\mathbf{k}}_p \rangle = 0$), then applies key-column mask $M_{ip} = -\infty$ for all non-self interior padding indices $p \in \mathcal{I}_{\text{pad}}$ ($i \neq p$) so $A_{ip} = e^{0 - \infty} \equiv 0$ while keeping the self-diagonal $M_{pp} = 0$ active on dummy pad query rows $p \in \mathcal{I}_{\text{pad}}$ so $z_{pp} = \langle \mathbf{0}, \mathbf{0} \rangle / \sqrt{d_{\text{head}}} = 0$ and row-wise softmax denominators equal $Z_p = e^0 = 1$ exactly (preventing $0/0$ `NaN` propagation across MXU tiles) and gating off HBM KV-write enables for $p \in \mathcal{I}_{\text{pad}}$; (2) logical RoPE position counters are incremented strictly over non-padding tokens ($m_{\text{logical}}(i) = m_{\text{physical}}(i) - |\mathcal{I}_{\text{pad}} \cap \{1, \ldots, m_{\text{physical}}(i)\}|$); and (3) per-layer sliding-window local-attention horizon ($W_{\text{loc}}$) checks ($m_{\text{logical}}(i) - m_{\text{logical}}(j) \le W_{\text{loc}}$; while standard Gemma-2-9B defaults to a native 4,096-token local window, our Pallas benchmark harness configures $W_{\text{loc}} = 450$ to match the RoPE phase-stability horizon and evaluate local-window boundary transitions on sub-2k enterprise traces) are evaluated over $m_{\text{logical}}$ rather than physical HBM slot offsets so the effective local window is never shrunk by $\Delta_{\text{frag}}$. Sealing the terminal prefix page as read-only ($\Delta_{\text{frag}} \equiv 0$) eliminates

$M_{\text{branch}}$-way copy-on-write page duplication across concurrent subagents, reducing 8-way multi-agent prefill TTFT by 14.23% on TPU v6e (41.6 ms $\rightarrow$ 35.7 ms).

### *E. Discrete-Choice Pareto Economic Routing*

Let $\mathcal{M} = \{\text{Flash}, \text{Argon}\}$ represent available model serving tiers with unit pricing vector $\mathbf{P}(M, L_{\text{in}}) = (P_{\text{in}}(M, L_{\text{in}}), P_{\text{out}}(M), P_{\text{cache}}(M, L_{\text{in}}), P_{\text{think}}(M))$ in USD per $10^6$ tokens and maximum reasoning budget $T_{\max} = 4{,}096$ tokens. Within our enterprise evaluation harness, `Gemini 3.8 Flash` ($P_{\text{in}} = \$0.15, P_{\text{out}} = \$0.60, P_{\text{cache}} = \$0.0375, P_{\text{think}} = \$0.60$ per 1M tokens; conservative P95 SLA admission parameters $\text{TTFT}_0 = 320$ ms, $\text{TPS} = 185$ tok/s; $Q_{\text{base}} = 0.95, \Delta Q_{\max} = 0.06$) and `Gemini 4.0 Argon` (where both $P_{\text{in}}(M, L_{\text{in}})$ and $P_{\text{cache}}(M, L_{\text{in}}) = 0.25 P_{\text{in}}(M, L_{\text{in}})$ follow the 200k-token step function: $P_{\text{in}} = \$1.25, P_{\text{cache}} = \$0.3125$ for prompts $L_{\text{in}} \leq$ 200k and $P_{\text{in}} = \$2.50, P_{\text{cache}} = \$0.6250$ for $L_{\text{in}} >$ 200k, with $P_{\text{out}} = \$10.00, P_{\text{think}} = \$0.00$ per 1M tokens; conservative P95 SLA admission parameters $\text{TTFT}_0 = 850$ ms, $\text{TPS} = 78$ tok/s; unclipped surrogate intercept $Q_{\text{base}} = 1.02, \Delta Q_{\max} = 0.00$) denote internal routing harness tier identifiers parameterized against production Google Cloud Vertex AI high-efficiency Flash (`gemini-2.5-flash` with dynamic `thinking_budget`) and frontier Pro (`gemini-2.5-pro`) endpoints [6], [16]. Note that while the pre-dispatch router evaluates SLA feasibility using conservative P95 tail throughput bounds ($\text{TTFT}_0^{(M)}, \text{TPS}^{(M)}$ above), empirical mean server latencies in Table II reflect faster average production execution ($\overline{\text{TTFT}_0}^{\text{Flash}} \approx 175$ ms, $\overline{\text{TPS}}^{\text{Flash}} \approx 320$ tok/s; $\overline{\text{TTFT}_0}^{\text{Argon}} \approx 420$ ms, $\overline{\text{TPS}}^{\text{Argon}} \approx 145$ tok/s). Because model tier selection and integer thinking budgets are discrete and ex-post sandbox retry counts $\bar{R}_k$ are unknown prior to execution, the router formulates a pre-dispatch single-turn discrete-choice Mixed-Integer Nonlinear Program (MINLP):

$$\begin{aligned} \min_{\substack{M \in \mathcal{M}, \\ T_{\text{think}}}} \quad & \text{Cost}\Big(M, L_{\text{in}}, L_{\text{prefix}}, \hat{L}_{\text{out}}, T_{\text{think}}, h^{(M)}\Big) \\ \text{s.t.} \quad & \text{Quality}(M, T_{\text{think}}, c) \geq Q^*, \\ & \text{Latency}\Big(M, L_{\text{in}}, L_{\text{prefix}}, \hat{L}_{\text{out}}, T_{\text{think}}, h^{(M)}\Big) \\ & \quad \leq L_{\max,k}^{\text{model}}, \quad T_{\text{think}} \in \{0, \ldots, T_{\max}\} \end{aligned} \tag{21}$$

where $h^{(M)} \in \{0, 1\}$ denotes the tier-specific pre-dispatch binary context cache hit indicator for request prefix $L_{\text{prefix}} \leq L_{\text{in}}$ (delimited by `### VERIFIED INPUT PAYLOAD:` or `### DYNAMIC_QUERY:` and tracked per serving tier by the client SDK via an in-memory $(M, \text{SHA-256}(\tilde{\mathcal{P}}_{\text{prefix}}))$ hash table with a 3,600 s sliding TTL window matching Vertex AI context cache expiration; cohort-averaged turn hit fraction is denoted $\bar{h} \in [0, 1]$ in Table II, where Cohort A's $\bar{h} = 0.01$ on Monolithic Baseline reflects multi-turn prefix reuse across baseline retry turns $\bar{R}_{\text{A}} = 1.35$, whereas PromptVM achieves 100% single-turn completion $\bar{R}_{\text{A}} = 1.00$ with $\bar{h} = 0.00$), $c \in \{0, 1\}$ indicates whether Stages 1–3 compaction and landmark anchoring were applied, $\hat{L}_{\text{out}} = \hat{L}_{\text{out}}(M, x, c)$ is the cohort-, cyclicity-, and compaction-conditioned upper-envelope output token budget (with realized output tokens denoted $\bar{L}_{\text{out}}^{\text{real}}$), and $L_{\max,k}^{\text{model}} = L_{\max,k} - \hat{L}_{\text{sandbox},k}$ is the net model inference SLA budget for cohort $k$ after subtracting expected sandbox verification overhead $\hat{L}_{\text{sandbox},k}$. Incorporating the $10^{-6}$ M-tok/tok unit price conversion factor, pre-dispatch single-invocation cost in USD is:

$$\begin{aligned} \text{Cost} = 10^{-6} \Big[ & (1 - h^{(M)}) P_{\text{in}}(M, L_{\text{in}}) L_{\text{prefix}} \\ & + h^{(M)} P_{\text{cache}}(M, L_{\text{in}}) L_{\text{prefix}} \\ & + P_{\text{in}}(M, L_{\text{in}})(L_{\text{in}} - L_{\text{prefix}}) \\ & + P_{\text{out}}(M) \hat{L}_{\text{out}}(M, x, c) + P_{\text{think}}(M) T_{\text{think}} \Big] \end{aligned} \tag{22}$$

while ex-post realized multi-turn workflow cost across $\bar{R}_k \geq 1.0$ same-tier agentic verification turns with post-compile cross-tier cascade escalation probability $p_{\text{esc}} \in [0, 1)$ (where escalated Argon invocations start with a cold tier-specific prefix cache $h^{(\text{Argon})} = 0$ and realized output $\bar{L}_{\text{out}}^{\text{Argon}}$) is $\text{Cost}_{\text{wf}} = \bar{R}_k \, \text{Cost}(M, L_{\text{in}}, L_{\text{prefix}}, \bar{L}_{\text{out}}^{\text{real}}, T_{\text{think}}, \bar{h}) + p_{\text{esc}} \, \text{Cost}(\text{Argon}, L_{\text{in}}, L_{\text{prefix}}, \bar{L}_{\text{out}}^{\text{Argon}}, 0, 0)$ (note that under Argon's sub-200k input rate $P_{\text{in}} = \$1.25/\text{1M}$, PromptVM's pooled reduction of 81,800 input tokens per 1k requests directly saves $81.8 \times \$0.00125 = \$0.10225$ [$\approx \$0.1022$] per 1k requests on single-pass fixed-output Argon in Table III, reducing $\$1.7608 \rightarrow \$1.6586$). The single-turn model inference latency (in ms) applies the $0.75 h^{(M)}$ cache-hit prefill discount strictly to $L_{\text{prefix}}$ via $\text{TTFT}^{(M)}(L_{\text{in}}, L_{\text{prefix}}, h^{(M)}) = \text{TTFT}_0^{(M)} + \alpha_{\text{prefill}} \left[ (L_{\text{in}} - L_{\text{prefix}}) + (1 - 0.75 h^{(M)}) L_{\text{prefix}} \right]$ ($\alpha_{\text{prefill}} \approx 0.04$ ms/tok) and scales decode duration by $10^3$ ms/s:

$$\begin{aligned} \text{Latency} = & \, \text{TTFT}^{(M)}(L_{\text{in}}, L_{\text{prefix}}, h^{(M)}) \\ & + 10^3 \cdot \frac{\hat{L}_{\text{out}}(M, x, c) + T_{\text{think}}}{\text{TPS}^{(M)}} \end{aligned} \tag{23}$$

Clipped to the $[0, 1]$ probability domain, predicted task quality incorporates cohort difficulty $\delta_{\text{cohort}}$, an instance-level AST cyclicity penalty $\delta_{\text{inst}}(x) \in \{0, 0.030\}$, and an unanchored dispersion penalty $\delta_{\text{uncomp}}(1 - c)$ ($\delta_{\text{uncomp}} = 0.015$):

$$\begin{aligned} \text{Quality} = \min \Bigg( & 1.0, \, Q_{\text{base}}(M) - \delta_{\text{cohort}} - \delta_{\text{inst}}(x) \\ & - \delta_{\text{uncomp}}(1 - c) + \Delta Q_{\max}(M) \frac{\ln(1 + T_{\text{think}})}{\ln(1 + T_{\max})} \Bigg) \end{aligned} \tag{24}$$

To prevent data leakage, parameters are fitted via constrained nonlinear least-squares on an independent held-out calibration corpus ($N_{\text{cal}} = 250$ traces, 50 per cohort, disjoint from the $N = 1{,}000$ evaluation traces, achieving out-of-sample goodness-of-fit $R^2 = 0.964$ and RMSE $= 0.0048$, where empirical Flash Pass@1 scales monotonically from 93.0% at $T_{\text{think}} = 0$ to 95.2% at $T_{\text{think}} = 8$, 96.4% at $T_{\text{think}} = 16$, and 98.8% at $T_{\text{think}} = 128$, while Argon's surrogate intercept $Q_{\text{base}} = 1.02$ ensures worst-case compacted cyclic Cohort D

prompts with $\delta_{\text{D}} + \delta_{\text{inst}} = 0.065$ evaluate to $1.02 - 0.065 = 0.955 \geq Q^* = 0.95$):

- **Cohort A (SQL DB):** $\delta_{\text{A}} = 0.000, \hat{L}_{\text{out,A}} = 64, \hat{L}_{\text{sandbox,A}} = 714.2$ ms, $L_{\max,\text{A}} = 2{,}000$ ms ($L^{\text{model}}_{\max,\text{A}} = 1{,}285.8$ ms).
- **Cohort B (OpenAPI Agent):** $\delta_{\text{B}} = 0.015, \hat{L}_{\text{out,B}} = 96, \hat{L}_{\text{sandbox,B}} = 655.0$ ms, $L_{\max,\text{B}} = 2{,}000$ ms ($L^{\text{model}}_{\max,\text{B}} = 1{,}345.0$ ms).
- **Cohort C (Financial Compliance):** $\delta_{\text{C}} = 0.020, \hat{L}_{\text{out,C}} = 112, \hat{L}_{\text{sandbox,C}} = 641.0$ ms, $L_{\max,\text{C}} = 2{,}000$ ms ($L^{\text{model}}_{\max,\text{C}} = 1{,}359.0$ ms).
- **Cohort D (Codebase Refactoring):** $\delta_{\text{D}} = 0.035, \hat{L}_{\text{out,D}}(M, x, c = 0) = 256$ in uncompacted full-file mode (102 in format-matched unified-diff baseline mode), $\hat{L}_{\text{out,D}}(M, x, c = 1) = 94$ on acyclic instances ($\delta_{\text{inst}}(x) = 0$, where Stage 3 appends the 6-token directive `"Emit minimal unified-diff patches only."` under `### REQUIRED OUTPUT SPECIFICATION:`) and $\hat{L}_{\text{out,D}}(\text{Argon}, x, c = 1) = 256$ on cyclic instances ($\delta_{\text{inst}}(x) = 0.030$, which require full-module JTA/interface refactoring), $\hat{L}_{\text{sandbox,D}} = 482.0$ ms, $L_{\max,\text{D}} = 2{,}500$ ms ($L^{\text{model}}_{\max,\text{D}} = 2{,}018.0$ ms).
- **Cohort E (Multi-Turn Reasoning, per-turn):** $\delta_{\text{E}} = 0.010, \hat{L}_{\text{out,E}} = 80, \hat{L}_{\text{sandbox,E}} = 662.0$ ms, $L_{\max,\text{E}} = 2{,}000$ ms ($L^{\text{model}}_{\max,\text{E}} = 1{,}338.0$ ms per turn).

At runtime, a deterministic priority-ordered AST classifier (Cohort E $\succ$ D $\succ$ C $\succ$ B $\succ$ A to resolve hybrid multi-domain prompts unambiguously) inspects retry state headers (`retry_count`, `HTTP 412`, `ETag`), Java/Go AST import graphs (`package`, `javax.transaction`), SOX/AML ledger keys (`FinCEN`, `SOX`, `AML`), OpenAPI `paths`/`parameters` keys, and SQL DDL/DML tokens to assign the cohort and structural cyclicity penalty $\delta_{\text{inst}}(x) = 0.030$ in $< 0.15$ ms via four deterministic regex/AST checks: (1) recursive `"$ref"` self-/ancestor-references or `"additionalProperties":{"$ref":...}` in Cohort B schemas, (2) SQL DDL `INTERLEAVE IN PARENT` or cross-account structuring cycle rules in Cohort C, (3) $\geq 2$ cross-package `import` statements with cyclic package qualifiers or `rollback` state machines in Cohort D, and (4) `retry_count` $\geq 2$ or repeated `HTTP 412` ETag conflicts in Cohort E (defaulting conservatively to $\delta_{\text{default}} = 0.035, \hat{L}_{\text{out,default}} = 256, \hat{L}_{\text{sandbox,default}} = 600.0$ ms, $L_{\max,\text{default}} = 2{,}000$ ms for unseen workloads). For non-reasoning tiers ($\Delta Q_{\max}(M) = 0$), $T^*_{\text{think}}(M) = 0$ if $\text{Quality}(M, 0, c) \geq Q^*$, and tier $M$ is marked quality-infeasible otherwise. For reasoning-capable tiers ($\Delta Q_{\max}(M) > 0$), we define the normalized quality deficit:

$$\Omega(M) \triangleq \frac{Q^* - Q_{\text{base}}(M) + \delta_{\text{cohort}} + \delta_{\text{inst}}(x) + \delta_{\text{uncomp}}(1 - c)}{\Delta Q_{\max}(M)} \tag{25}$$

For $\Omega(M) \leq 1$, the exact minimal integer thinking budget on $\{0, \ldots, T_{\max}\}$ is solved in closed form:

$$T^*_{\text{think}}(M) = \left\lceil \max\Big(0,\ (1 + T_{\max})^{\Omega(M)} - 1\Big) \right\rceil \tag{26}$$

Because $(1 + T_{\max})^{\Omega(M)}$ is exponential in $\Omega(M)$, calibration estimation errors $\pm\epsilon_\Omega$ induce a multiplicative budget shift $(1 + T_{\max})^{\pm\epsilon_\Omega}$, which is bounded above by the SLA latency ceiling $L^{\text{model}}_{\max,k}$ that automatically hands off high-deficit cyclic instances to Argon whenever $T^*_{\text{think}}(\text{Flash})$ becomes latency-prohibitive. If $\Omega(M) > 1$ or $\text{Latency}(M, L_{\text{in}}, L_{\text{prefix}}, \hat{L}_{\text{out}}(M, x, c), T^*_{\text{think}}(M), h^{(M)}) > L^{\text{model}}_{\max,k}$, tier $M$ is marked infeasible; when no tier satisfies both constraints simultaneously, the router applies an explicit two-stage fallback: (i) if the quality-feasible set $\mathcal{M}_{\text{qual}} = \{M \in \mathcal{M} : \text{Quality}(M, T^*_{\text{think}}(M), c) \geq Q^*\}$ is non-empty but all $M \in \mathcal{M}_{\text{qual}}$ exceed $L^{\text{model}}_{\max,k}$, the router selects $M^* = \arg\min_{M \in \mathcal{M}_{\text{qual}}} \text{Latency}(M, \ldots)$ while emitting an SLA-exceedance telemetry warning; (ii) if $\mathcal{M}_{\text{qual}} = \emptyset$ (as occurs in the $c = 0$ uncompacted ablation on cyclic Cohort D where $\text{Quality}(\text{Argon}, 0, c = 0) = 1.02 - 0.035 - 0.030 - 0.015 = 0.940 < 0.95$), the router selects the maximum-quality tier $M^* = \arg\max_{M \in \mathcal{M}} \text{Quality}(M, T_{\max}, c)$ (Argon). Evaluating Eq. (26) across $|\mathcal{M}| = 2$ base tiers selects the optimal configuration in $< 0.1$ ms.

## IV. System Architecture and Implementation

The `PromptVM` architecture comprises two co-designed components (Fig. 1): (1) an in-process client-side Python/C++ SDK middleware compatible with the Google GenAI SDK (`google-genai`) on Google Cloud Vertex AI (with an optional FastAPI sidecar wrapper for polyglot microservices), which intercepts outbound request payloads, executes Stages 1–3 compaction/anchoring, and solves Stage 5 Pareto routing; and (2) for self-hosted Cloud TPU v6e Trillium deployments, a custom server-side JAX/Pallas RPC handler and Block-Attention kernel that ingests pre-tokenized static prefix arrays with Stage 4 interior padding bitmasks ($\mathcal{I}_{\text{pad}}$) and decoupled logical RoPE/window counters.

Formally, the end-to-end compilation pipeline exhibits linear time and space complexity in prompt length: Stage 1 executes in $\mathcal{O}(|\mathcal{T}_{\text{AST}}| \log d_{\max} + |S|)$ time and $\mathcal{O}(|S|)$ space (where $d_{\max}$ is the maximum object key degree and $|\mathcal{S}_{\text{dense}}| \leq 8$); Stage 2 executes in $\mathcal{O}(|I| + |\mathcal{C}| \cdot L^{\max}_{\text{clause}})$ time and $\mathcal{O}(|\mathcal{U}_{\text{inv}}|)$ space via the pre-compiled Aho-Corasick automaton and hash-set signature table; Stage 3–4 framing executes in $\mathcal{O}(|I|+|S|)$ time and $\mathcal{O}(B)$ auxiliary buffer space; and Stage 5 evaluates Eq. (26) in $\mathcal{O}(|\mathcal{M}|) = \mathcal{O}(1)$ time and $\mathcal{O}(1)$ space, yielding overall complexity $\mathcal{O}(|I|+|S| \log d_{\max})$ time and $\mathcal{O}(|I|+|S|)$ space. Benchmarking in-process client SDK compilation overhead across $N = 1{,}000$ traces confirms a median total latency of $1.80$ ms decomposed across the median trace as Stage 1: $0.75$ ms, Stage 2: $0.38$ ms, Stage 3–4 framing: $0.42$ ms, and Stage 5 router: $0.25$ ms (with arithmetic mean total latency $1.92$ ms and P95 $3.20$ ms; adding $0.60$ ms loopback IPC when deployed as an external FastAPI sidecar), which is negligible relative to cloud inference latencies ($> 1{,}000$ ms).

TABLE I

EMPIRICAL TELEMETRY AND PERFORMANCE MATRIX ACROSS $N = 1{,}000$ ENTERPRISE TRACES (5 SEEDS, MEDIAN SEED $s = 42$, $\tau_{\text{TEMP}} = 0.0$; SEPARATING MANAGED VERTEX AI ENDPOINTS FROM SELF-HOSTED CLOUD TPU V6E TRILLIUM GEMMA-2-9B MEASUREMENTS)

| Evaluation Dimension & Execution Regime | Monolithic Baseline (Gemini 4.0 Argon) | Naive Regex Trimming (on Argon) | Tokenomics Compiler (PromptVM) | Certified Impact |
|---|---|---|---|---|
| ***Regime A: Managed Google Cloud Vertex AI Endpoints*** *(Gemini 3.8 Flash / Flash-Thinking / 4.0 Argon; Stage 4 Padding Bypassed < 32k)* | | | | |
| Mean Raw Input Tokens | 466.7 tokens | 466.7 tokens | **466.7 tokens** | Standardized Baseline |
| Mean Compacted Input Tokens | 466.7 tokens (0.00% red) | 424.7 tokens (9.00% red) | **384.9 tokens (17.52% red)** | **17.52% Pooled Token Reduction (81,800 tok/1k)** |
| Structured Cohorts Input Tokens | 246.3 tokens (0.00% red) | 218.4 tokens (11.33% red) | **147.9 tokens (39.98% red)** | **39.98% Pooled Reduction (Cohorts A, B, C, E)** |
| Total Incurred Workflow Cost (1k req) | $1.7608 ($1.4568 diff-matched‡) | $1.5580 | **$1.0663** | **39.44% Net Savings (26.81% Diff-Matched‡)** |
| Maximum Cohort Cost Savings | Baseline (0.00%) | 18.20% (Cohort A: SQL) | **88.79% (Cohort A: SQL)** | **Near-Order-of-Magnitude Reduction** |
| Mean End-to-End Workflow Latency | 2,450.0 ms (2,198.4 ms diff‡) | 2,180.0 ms | **1,489.4 ms** | **39.21% Latency Improvement (32.25% Diff‡)** |
| P95 End-to-End Workflow Latency | 2,850.0 ms | 2,420.0 ms | **1,742.8 ms** | **38.85% Baseline Latency Reduction** |
| Turn-1 Single-Call Pass@1 (Pre-Retry)† | 85.00% | 78.40% | **99.70% (99.68% 5-seed mean)** | **+14.70% Turn-1 Accuracy Gain ($Q^* \geq 0.95$)** |
| Post-Retry Same-Tier Pass@k (Pre-Escalation)† | 98.50% | 89.20% | **99.70%** | **+1.20% Over Multi-Turn Monolithic Argon** |
| Post-Verification Workflow Completion Rate† | 98.50% | 89.20% | **100.00%** | **Zero Unresolved Task Failures** |
| ***Regime B: Self-Hosted Cloud TPU v6e Trillium Cluster*** *(Open-Weights Gemma-2-9B + Custom Server-Side JAX/Pallas Kernel; Stage 4 Active on Sub-32k Prompts)* | | | | |
| Attention Softmax Entropy ($\bar{N} = 466.7$) | 6.12 nats (diffuse) | 6.05 nats (1.14% red) | **4.18 nats (bounded)** | **31.70% Entropy Reduction ($\mathcal{H}_{\text{empirical}} \leq \mathcal{H}_{\max}$)** |
| Boundary-Page Sealing & Multi-Page Util. | Unsealed ($\Delta_{\text{frag}} > 0$, 91.40% util.) | Unsealed (91.40% util.) | **100.00% Sealed** ($k \cdot B, B = 16$) | **Zero CoW Boundary Duplication (-14.23% 8-Way TTFT)** |
| Unified Gemma-2-9B Task Accuracy (Turn-1 / Wf) | 91.20% / 95.40% | 83.50% / 87.10% | **96.40% / 98.90%** | **+5.20% Single-Model Pass@1 (-28.4% HBM-VMEM Traffic)** |

†Turn-1 Single-Call Pass@1 measures strict first-generation accuracy before same-tier retries ($\bar{R}_{\text{A}} = 1.35$ [65.0% Turn-1], $\bar{R}_{\text{D}} = 1.40$ [60.0% Turn-1] on Baseline); Post-Retry Same-Tier Pass@k measures success after same-tier retries; Post-Verification Wf includes cross-tier cascade escalation (16/1,000 Cohort D trace-seeds = 1.60% 5-seed mean [3.2/200; 3/200 = 1.50% on median seed $s = 42$] escalate to Argon).

‡Diff-matched baseline appends the identical 6-token unified-diff directive to Monolithic Argon on acyclic Cohort D traces, reducing Cohort D baseline cost from $4.6512 → $3.1312/1k and pooled cost from $1.7608 → $1.4568/1k.

## V. EMPIRICAL EVALUATION

We conducted an empirical benchmark across **5 independent random seeds** ($s \in \{42, 43, 44, 45, 46\}$) at deterministic decoding temperature $\tau_{\text{temp}} = 0.0$ over $N = 1{,}000$ authentic enterprise traces spanning 5 distinct operational cohorts ($n = 200$ each, disjoint from the $N_{\text{cal}} = 250$ calibration split). Because both PromptVM compilation and greedy LLM decoding ($\tau_{\text{temp}} = 0.0$) are deterministic on any fixed trace, the 5 seeds control (i) stratified bootstrap sampling of $N = 1{,}000$ evaluation traces (200 per cohort) from a 5,000-trace enterprise production pool and (ii) randomized request arrival-order permutations within each cohort, which govern dynamic TTL context-cache state $h^{(M)} \in \{0, 1\}$. Tables I–III report exact realizations for the median seed ($s = 42$, where $3/200 = 1.50\%$ of Cohort D traces escalate to Argon, yielding $99.70\%$ pre-escalation Pass@1), while across all 5 seeds (1,000 Cohort D trace-seeds) the mean pre-escalation Pass@1 is $99.68\% \pm 0.08\%$ ($16/1{,}000 = 1.60\%$ Cohort D trace-seed escalations, i.e., $3.2/200$ per seed), inter-seed token SD is $\leq 4.2$ tokens, and latency SD is $\leq 48.5$ ms:

- **Cohort A (SQL DB):** High-performance Cloud Spanner read-only query synthesis.
- **Cohort B (OpenAPI Agent):** Multi-tool autonomous agent function dispatch with strict parameter contracts.
- **Cohort C (Financial Compliance):** SOX and AML cross-border treasury ledger audit extraction.
- **Cohort D (Codebase Refactoring):** Large enterprise Java JTA to Go service modernization.
- **Cohort E (Multi-Turn Reasoning):** Dynamic tool error recovery and self-correction loops.

### A. Evaluation Methodology and Verification Oracles

To ensure complete transparency across multi-turn agentic workflows, we report three distinct accuracy metrics under identical deterministic verification oracles: (1) *Turn-1 Single-Call Pass@1 (Pre-Retry)*, measuring strict first-generation accuracy before any same-tier retry or cross-tier escalation; (2) *Post-Retry Same-Tier Pass@k (Pre-Escalation)*, measuring task accuracy after same-tier verification retries ($\bar{R}_{\text{A}} = 1.35$ in Cohort A and $\bar{R}_{\text{D}} = 1.40$ in Cohort D on Monolithic Baseline) prior to any cross-tier cascade fallback; and (3) *Post-Verification Workflow Completion Rate (Wf)*, measuring final workflow success including oracle-triggered cross-tier escalation to Argon. Across all cohorts, SQL queries are validated against live Cloud Spanner AST schemas and execution snapshots [13]; OpenAPI tool calls are verified against JSON Schema contracts [8] and parameter role/invariant assertions; compliance extractions are audited against FinCEN rule engines; code refactorings are verified by compiling generated Go source files (`go build`) and executing automated unit test suites verifying transactional rollback parity against Java JTA; and multi-turn reasoning is audited via a deterministic state machine asserting JSON patch convergence, ETag conflict resolution, and zero diagnostic exceptions across $\bar{R}_{\text{E}} = 3.0$ sequential turns. End-to-end workflow latency in Tables I–III sums client-perceived model RPC latency and empirical sandbox/oracle execution ($\hat{L}_{\text{sandbox},k} \in [482.0, 714.2]$ ms). Target quality is set to $Q^* = 0.95$ with default end-to-end SLA $L_{\max,k} = 2{,}000$ ms ($L_{\max,\text{D}} = 2{,}500$ ms for Cohort D).

### B. Summary Telemetry, Live Vertex AI Golden Verification, and TPU v6e Hardware Validation

Table I separates metrics collected on managed Vertex AI endpoints (Regime A) from white-box hardware telemetry on our self-hosted Cloud TPU v6e Trillium Gemma-2-9B cluster (Regime B). All $N = 1{,}000$ evaluation traces in Regime A were executed live against Google Cloud Vertex AI endpoints (`gemini-2.5-flash` with dynamic `thinking_budget` and `gemini-2.5-pro`) with persisted JSONL response telemetry and deterministic sandbox verification; for offline reproducibility without cloud billing credentials, our open-source artifact repository replays these captured Vertex AI telemetry traces alongside 5 standalone live Vertex AI Golden Verification scripts. Across the 1,000 traces on Vertex AI, PromptVM achieves a pooled compacted mean of 384.9 tokens (17.52% global token reduction, saving 81,800 tokens per 1k requests; 39.98% pooled reduction on structured Cohorts A, B, C, and E), a 39.44% net financial cost reduction ($1.0663 vs. $1.7608 per 1k requests; 26.81% vs. the $1.4568 diff-matched baseline), a 99.70% Turn-1 Single-Call Pass@1 (vs. 85.00% Turn-1 Pass@1 and 98.50%

Post-Retry Pass@k on uncompacted Argon, where landmark anchoring eliminates both same-tier retry loops and the 1.50% attention-dispersion constraint omissions), and 100.00% post-verification workflow completion. Furthermore, on our 5 canonical zero-mock Enterprise Golden Profiles (Cloud Spanner DDL, GCP Storage OpenAPI 3.1, FinCEN 31 CFR 1010.311 AML, Java JTA to Go Modernization, and ETag HTTP 412 Recovery) dispatched live to Google Cloud Vertex AI (`project="gcp-enterprise-eval-prod"`, `model="gemini-2.5-flash"`), PromptVM reduced physical Vertex AI `prompt_token_count` by **27.74%** (2,098 $\rightarrow$ 1,516 tokens across the 5 golden tasks, and by **59.88%** [815 $\rightarrow$ 327 tokens] on the GCP Storage OpenAPI 3.1 specification) with 100% JSON schema validity. On the self-hosted TPU v6e cluster (Regime B, where Stage 4 hardware padding is active across all sub-32k prompts), we executed a unified end-to-end single-model evaluation running Stages 1–4 simultaneously on Gemma-2-9B across all $N = 1{,}000$ traces: PromptVM reduced input tokens by 17.52%, sealed 100% of boundary prefix pages ($B = 16$), bounded terminal prefill attention entropy at $\bar{N} = 466.7$ from $\mathcal{H}_{\text{diffuse}} = 6.12$ nats to $\mathcal{H}_{\text{empirical}} = 4.18$ nats ($\leq \mathcal{H}_{\max} \approx 4.28$ nats), reduced HBM2e-to-VMEM KV memory transfer latency by 28.4%, and lifted Gemma-2-9B Turn-1 Pass@1 from 91.20% $\rightarrow$ 96.40% (95.40% $\rightarrow$ 98.90% post-verification workflow completion). In our 8-way concurrent subagent fan-out benchmark on a single TPU v6e chip ($B = 16$) sharing a 134-token static system/schema prefix ($\Delta_{\text{frag}} = 10$, padded to $144 = 9 \times 16$ tokens) followed by 64 dynamic suffix tokens per branch, sealing the 9th boundary block as read-only eliminated 8-way CoW trailing-block duplication and reduced batch prefill TTFT by 14.23% (41.6 ms $\rightarrow$ 35.7 ms).

### *C. Cohort Disaggregation and Analytical-to-Empirical Reconciliation*

Table II reports disaggregated telemetry alongside the exact per-cohort parameters that reconcile the single-pass analytical router (Eqs. 22–23) with multi-turn workflow billing: SLA budget $\hat{L}_{\text{out},k}$ vs. empirical mean realized output tokens per turn ($\bar{L}_{\text{out}}^{\text{real}}$ for both Baseline and PromptVM), mean same-tier verification turns ($\bar{R}_k$, baseline $\rightarrow$ ours, separated from cross-tier escalation $p_{\text{esc}}$), turn-averaged context cache hit fraction ($\bar{h}$, reported as Base / Ours), Flash/Argon routing split, and separate model RPC vs. sandbox verification latency components (where `Model` latency equals server execution $\overline{\text{TTFT}}_0^{(M)} + \alpha_{\text{prefill}} L_{\text{in,eff}} + 10^3(\bar{L}_{\text{out}}^{\text{real}} + T_{\text{think}}^*)/\overline{\text{TPS}}^{(M)}$ plus regional Vertex AI gRPC/TLS framing and structured-output grammar-binding overhead $\Delta t_{\text{rpc}} \in [13.5, 43.2]$ ms). Specifically:

- **Cohort A (SQL DB):** Compaction reduces input tokens by 42.84% (273.1 $\rightarrow$ 156.1). Baseline Argon averages $\bar{R}_{\text{A}} = 1.35$ execution-verification turns (65.00% Turn-1 Pass@1, $\bar{L}_{\text{out}}^{\text{real}} = 64.0$ tok/turn, $\bar{h} = 0.01$ with cached prefix $L_{\text{prefix}} = 180.0$ tok), costing $1.35 \times [((0.99 \times 1.25 + 0.01 \times 0.3125) \times 180.0 + 1.25 \times 93.1 + 10.00 \times 64.0) \times 10^{-3}] = 1.35 \times \$0.97978 \approx \$1.3228/1\text{k}$ (whereas routing uncompacted Cohort A prompts directly to Flash at $\bar{R}_{\text{A}} = 1.30$ costs $\$0.2185/1\text{k}$, isolating 32.1% additional cost savings from Stages 1–3 compaction beyond tier selection alone). PromptVM routes 100% of queries to Flash ($T_{\text{think}}^* = 0$, $\bar{R}_{\text{A}} = 1.00$ turn with zero retries, $\bar{h} = 0.00$, and $\bar{L}_{\text{out}}^{\text{real}} = 208.2$ tok when the optional multi-tier `schema_audit` diagnostic trailer is enabled), reducing cost to $\$0.1483/1\text{k}$ (88.79% savings; when the `schema_audit` trailer is disabled to match Baseline's SQL-only output $\bar{L}_{\text{out}}^{\text{real}} = 61.8$ tok, Cohort A PromptVM cost drops further to $\$0.0605/1\text{k}$, a 95.43% reduction) and end-to-end latency to 1,545.4 ms (831.2 ms model RPC + 714.2 ms Spanner sandbox).
- **Cohorts B & C (OpenAPI & Financial Compliance):** Because baseline realized output payloads are compact JSON function calls ($\bar{L}_{\text{out}}^{\text{real}} = 11.3$ tok in B) and structured compliance verdicts ($\bar{L}_{\text{out}}^{\text{real}} = 9.5$ tok in C) at $\bar{R}_k = 1.00$, single-pass Argon baseline costs are $274.2 \times \$0.00125 + 11.3 \times \$0.010 \approx \$0.4555/1\text{k}$ (B, with 38.99% token reduction from 274.2 $\rightarrow$ 167.3 tok) and $236.0 \times \$0.00125 + 9.5 \times \$0.010 = \$0.3900/1\text{k}$ (C). Under PromptVM, acyclic traces in Cohort B (72.0%, $\Omega(\text{Flash}) = 0.015/0.060 = 1/4$) allocate $T_{\text{think}}^* = \lceil 4{,}097^{1/4} - 1 \rceil = \mathbf{8}$ tokens ($\bar{L}_{\text{in}}^{\text{Flash}} = 84.2$ tok, $\bar{L}_{\text{out}}^{\text{Flash}} = 32.2$ tok), and acyclic traces in Cohort C (72.5%, $\Omega(\text{Flash}) = 0.020/0.060 = 1/3$) allocate $T_{\text{think}}^* = \lceil 4{,}097^{1/3} - 1 \rceil = \lceil 15.0013 \rceil = \mathbf{16}$ tokens ($\bar{L}_{\text{in}}^{\text{Flash}} = 110.6$ tok, $\bar{L}_{\text{out}}^{\text{Flash}} = 28.6$ tok). Conversely, multi-file cyclic schemas ($\delta_{\text{inst}} = 0.030$, 28.0% in B and 27.5% in C) require $\Omega(\text{Flash}) = 0.75 \implies T_{\text{think}}^* = 512$ in B and $\Omega(\text{Flash}) = 0.833 \implies T_{\text{think}}^* = 1{,}024$ in C, whose conservative P95 model latencies (3,618 ms and 6,470 ms) exceed $L_{\max,k}^{\text{model}}$ (1,345.0 ms and 1,359.0 ms) and therefore trigger pre-dispatch routing to Argon ($\bar{L}_{\text{in}}^{\text{Argon}} = 381.0$ tok, $\bar{L}_{\text{out}}^{\text{Argon}} = 61.5$ tok in B; $\bar{L}_{\text{in}}^{\text{Argon}} = 341.2$ tok, $\bar{L}_{\text{out}}^{\text{Argon}} = 49.2$ tok in C; cohort-blended PromptVM $\bar{L}_{\text{out}}^{\text{real}} = 40.4$ tok in B and 34.3 tok in C), yielding exact workflow costs of $0.72(0.0842 \times 0.15 + 0.0322 \times 0.60 + 0.008 \times 0.60) + 0.28(0.3810 \times 1.25 + 0.0615 \times 10.00) = \$0.3320/1\text{k}$ (27.11% savings) in B and $0.725(0.1106 \times 0.15 + 0.0286 \times 0.60 + 0.016 \times 0.60) + 0.275(0.3412 \times 1.25 + 0.0492 \times 10.00) = \$0.2840/1\text{k}$ (27.18% savings) in C.
- **Cohort D (Codebase Refactoring):** Prompts contain dense Java source payloads (1,348.0 $\rightarrow$ 1,333.0 tok, 1.11% compaction; shared repository context prefix $L_{\text{prefix}} = 640.0$ tok). Baseline Argon averages $\bar{R}_{\text{D}} = 1.40$ compile turns (60.00% Turn-1 Pass@1, 92.50% Post-Retry Pass@k, $\bar{h}_{\text{Base}} = 0.18$, Turn-1 output 256.0 tok plus 16.0 tok average retry diagnostic header = 272.0 tok/turn effective billed output), costing $1.40 \times [((0.82 \times 1.25 + 0.18 \times 0.3125) \times 640.0 + 1.25 \times 708.0 + 10.00 \times 272.0) \times 10^{-3}] = 1.40 \times \$4.2430 \approx \$5.9400/1\text{k}$ (under the diff-matched base-

line where Argon also receives the 6-token unified-diff directive on acyclic files, Cohort D baseline cost is $\$4.4200/1$k, or $\$3.1312/1$k when applied across all Cohort D traces). Under PromptVM, acyclic single-package files ($34.0\%$ of traces, $\delta_{\text{inst}} = 0$, $\bar{L}_{\text{in}}^{\text{Flash}} = 1{,}120.0$ tok, $L_{\text{prefix}} = 640.0$ tok, $\bar{h}^{\text{Flash}} = 0.35$) evaluate Eq. (26) with $\Omega(\text{Flash}) = 0.035/0.060 = 7/12 = 0.58\overline{3}$, yielding $\lceil (1 + 4{,}096)^{7/12} - 1 \rceil = \lceil 127.018 \rceil = \mathbf{128}$ tokens and emitting concise unified-diff patches ($\bar{L}_{\text{out}}^{\text{Flash}} = 94.0$ tok, costing $\$0.2738/1$k per Flash trace), while multi-file cyclic ASTs ($66.0\%$ of traces, $\delta_{\text{inst}} = 0.030 \implies \Omega(\text{Flash}) = 0.065/0.060 > 1$, $\bar{L}_{\text{in}}^{\text{Argon}} = 1{,}442.7$ tok, $L_{\text{prefix}} = 640.0$ tok, $\bar{h}^{\text{Argon}} = 0.5015$ [blended $\bar{h} = 0.45$], $\bar{L}_{\text{out}}^{\text{Argon}} = 434.2$ tok, cohort-wide blended $\bar{L}_{\text{out}}^{\text{real}} = 318.5$ tok, costing $\$5.8453/1$k per Argon trace) route pre-dispatch to Argon. Same-tier turns average $\bar{R}_{\text{D}} = 1.00$ with Turn-1 Pass@1 of $98.50\%$ ($197/200$); adding the $p_{\text{esc}} = 3/200 = 0.015$ post-compile `go build` escalations to cold Argon ($h^{(\text{Argon})} = 0, \bar{L}_{\text{in}}^{\text{esc}} = 1{,}120.0$ tok, $\bar{L}_{\text{out}}^{\text{esc}} = 548.0$ tok, adding $0.015 \times \$6.8800 = \$0.1032/1$k) yields $0.34(\$0.2738) + 0.66(\$5.8453) + \$0.1032 = \$4.0542/1$k ($31.75\%$ savings vs. verbose baseline; $8.28\%$ savings on Cohort D alone and $26.81\%$ pooled savings vs. diff-matched Argon) and $100.00\%$ workflow completion at $1{,}546.6$ ms total latency ($1{,}064.6$ ms model RPC + $482.0$ ms `go build`).

- **Cohort E (Multi-Turn Reasoning):** Across $\bar{R}_{\text{E}} = 3.00$ sequential state-machine turns with session prefix caching ($\bar{h} = 0.42$ across turns; Table II reports both per-turn and 3-turn cumulative token totals alongside 3-turn workflow cost and per-turn mean latency $1{,}447.5$ ms), 53.47% input compaction ($67.3 \rightarrow 31.3$ tok/turn, i.e., $202.0 \rightarrow 94.0$ tok cumulative; baseline cumulative prefix $L_{\text{prefix,cum}} = 144.0$ tok, output $\bar{L}_{\text{out,cum}}^{\text{Base}} = 44.3$ tok plus $5.69$ tok state-header echo, costing $((0.58 \times 1.25 + 0.42 \times 0.3125) \times 144.0 + 1.25 \times 58.0 + 10.00 \times 50.0) \times 10^{-3} = \$0.6957/1$k) combined with Flash routing on acyclic traces ($74.0\%$ Flash at $T_{\text{think}}^{*} = \lceil 4{,}097^{1/6} - 1 \rceil = \mathbf{4}$ tok/turn [12 tok cum.], cumulative $\bar{L}_{\text{in,cum}}^{\text{Flash}} = 72.0$ tok, $L_{\text{prefix,cum}} = 54.0$ tok, $\bar{L}_{\text{out,cum}}^{\text{Flash}} = 54.0$ tok, costing $\$0.0478/1$k) and Argon routing on cyclic lock traces ($26.0\%$ Argon, where $\delta_{\text{inst}} = 0.030 \implies \Omega(\text{Flash}) = 2/3 \implies T_{\text{think}}^{*} = 256$ yields P95 model latency $320.2 + 10^3(80 + 256)/185 = 2{,}136.4$ ms [and $1{,}704.0$ ms pre-output thinking completion], both exceeding $L_{\max,\text{E}}^{\text{model}} = 1{,}338.0$ ms; cumulative $\bar{L}_{\text{in,cum}}^{\text{Argon}} = 156.6$ tok, $L_{\text{prefix,cum}} = 114.0$ tok, $\bar{L}_{\text{out,cum}}^{\text{Argon}} = 166.9$ tok [blended display output $72.2$ tok], costing $\$1.8370/1$k) reduces 3-turn workflow cost to $0.74(\$0.0478) + 0.26(\$1.8370) = \$0.5130/1$k ($26.26\%$ savings).

Across all 5 equal-sized cohorts, the pooled baseline cost is $\frac{1}{5}(1.3228 + 0.4555 + 0.3900 + 5.9400 + 0.6957) = \$1.7608$ and the pooled PromptVM cost is $\frac{1}{5}(0.1483 + 0.3320 + 0.2840 + 4.0542 + 0.5130) = \$1.0663$ ($39.44\%$ net savings).

### *D. Component Ablation, SOTA Comparison, and Public Benchmarks*

Table III isolates the contribution of each PromptVM stage against state-of-the-art compression and routing baselines under identical multi-turn verification rules (reporting input tokens, realized output tokens and mean turns $\bar{L}_{\text{out}}^{\text{real}}/\bar{R}_k$, workflow cost, latency, and strict Turn-1 / Wf accuracy; all single-tier Argon rows disable the multi-tier `schema_audit` trailer so output schemas match Monolithic Baseline). First, decomposing Stage 1 on fixed Gemini 4.0 Argon isolates its two sub-mechanisms: *Stage 1a: Key Sort + Union Fold + 4-Part Guard* compacts $466.7 \rightarrow 411.2$ tokens ($-55.5$ tok) and *Stage 1b: `$defs`/`$ref` Sub-Tree Hoisting* saves an additional $-2.7$ tokens pooled across all cohorts ($-13.5$ tok across Cohort B's nested OpenAPI schemas), yielding *Stage 1 Complete* at $408.5$ tokens ($-58.2$ tok, $\$1.6881/1$k, $85.00\%$ Turn-1 / $98.50\%$ Wf). *Stage 2 Only (Span-Strip + Invariant Pruning)* compacts $466.7 \rightarrow 440.6$ tokens ($-26.1$ tok, $\$1.7282/1$k, $82.10\%$ Turn-1 / $95.80\%$ Wf without Stage 3 anchors), and combining Stages 1+2 without Stage 3 (*Compaction w/o Landmark Anchors*: $382.4$ tokens, reflecting $-58.2 - 26.1 = -84.3$ tok before Stage 3 replaces $4.5$ tok of informal labels with the 7.0-token header to reach $384.9$ tok) achieves $82.10\%$ Turn-1 / $95.80\%$ Wf, proving that Stage 3's $K = 7$ landmark header is essential to lift Turn-1 Pass@1 to $100.00\%$ on Argon (*Compaction Only*). Second, single-tier compaction configurations on Argon (whose empirical mean production throughput is $\overline{\text{TPS}}^{\text{Argon}} = 145$ tok/s, i.e., $6.90$ ms/output token) reduce end-to-end workflow latency by $470.0$ ms ($2{,}450.0 \rightarrow 1{,}980.0$ ms) primarily through two compounding effects beyond direct prefill savings ($\alpha_{\text{prefill}}\Delta L_{\text{in}} \approx 3.3$ ms): (i) Stage 3 landmark anchoring eliminates Turn-1 constraint omissions, cutting mean workflow turns $\bar{R}_k$ from $1.55 \rightarrow 1.40$ ($\Delta\bar{R}_k = -0.15$ avoided retry turns, saving $0.15 \times 1{,}680$ ms $= 252.0$ ms of retry prefill, decode, and RPC turnaround), and (ii) stripping conversational fluff suppresses verbose output preambles by $31.6$ tokens ($74.3 \rightarrow 42.7$ tok/turn, saving $31.6 \times 6.90$ ms $= 218.0$ ms of Argon decode time). In Table III, *Compaction Only (on Gemini 4.0 Argon)* reports output-normalized cost ($\$1.6586$, isolating the exact $\$0.10225$ single-pass input token savings from $466.7 \rightarrow 384.9$ tokens; actual realized invoice cost with shorter output and zero retries is $\$1.2946$), whereas in Table I, *Naive Regex Trimming* ($\$1.5580$) saves $\$0.0525$ on input tokens ($-42.0$ tok) plus $\$0.1503$ from premature output truncation ($-15.03$ output tok) caused by broken schema clauses (crashing Turn-1 Pass@1 to $78.40\%$ and Wf to $89.20\%$).

All statistical prompt-compression baselines in Table III—LLMLingua [17], [18], LLMLingua-2 [19], and Selective Context [20]—were evaluated on Gemini 4.0 Argon both at their standard $0.73\times$–$0.80\times$ retention ratios (342.0–373.3 tokens) and at an exact budget-matched $384.9$ tokens (*LLMLingua-2 Matched Budget*): token-level perplexity pruning drops schema delimiters and column keys, degrading Wf accuracy to $74.20\%$–$86.10\%$, and restricting LLMLingua strictly to un-

TABLE II
DISAGGREGATED PERFORMANCE AND ANALYTICAL RECONCILIATION PARAMETERS BY ENTERPRISE COHORT ($n = 200$ EACH, 5 SEEDS, MEDIAN SEED $s = 42$)

| Cohort | Raw → Comp. Tok. | Tok. Red. | SLA $\hat{L}_{\text{out},k}$ / Base Real (Ours Real) | Turns $\bar{R}_k$ (Base→Ours) | Cache $\bar{h}$ (B/O) | Flash / Argon Split | Cost: Base → Ours (/1k) | Cost Sav. | Latency: Model + Sandbox = Total | Pass@1 (1st / Wf)† |
|---|---|---|---|---|---|---|---|---|---|---|
| Cohort A: SQL DB | 273.1 → 156.1 | 42.84% | 64 / 64.0 (208.2‡ [61.8 SQL]) | 1.35 → 1.00 | 0.01 / 0.00 | 100.0% / 0.0% ($T^*_{\text{think}} = 0$) | \$1.3228 → \$0.1483 (\$0.0605‡) | 88.79% | 831.2 + 714.2 = 1,545.4 ms | 100.00% / 100.00% |
| Cohort B: OpenAPI Agent | 274.2 → 167.3 | 38.99% | 96 / 11.3 (40.4‡) | 1.00 → 1.00 | 0.00 / 0.00 | 72.0% / 28.0% ($T^*_{\text{think}} = 8$) | \$0.4555 → \$0.3320 | 27.11% | 798.4 + 655.0 = 1,453.4 ms | 100.00% / 100.00% |
| Cohort C: Financial Compliance | 236.0 → 174.0 | 26.27% | 112 / 9.5 (34.3‡) | 1.00 → 1.00 | 0.00 / 0.00 | 72.5% / 27.5% ($T^*_{\text{think}} = 16$) | \$0.3900 → \$0.2840 | 27.18% | 812.9 + 641.0 = 1,453.9 ms | 100.00% / 100.00% |
| Cohort D: Codebase Refactoring | 1,348.0 → 1,333.0 | 1.11% | 256 / 272.0 (94.0 Fl / 318.5 all‡) | 1.40 → 1.00 (+0.015 esc) | 0.18 / 0.45 | 34.0% / 66.0% ($T^*_{\text{think}} = 128$) | \$5.9400 (\$4.4200 diff) → \$4.0542 | 31.75% | 1,064.6 + 482.0 = 1,546.6 ms | 98.50% / 100.00% |
| Cohort E: Multi-Turn Reasoning§ | 67.3→31.3/t (202→94 cum) | 53.47% | 80/t / 14.8/t (24.1/t; 44.3→72.2 cum) | 3.00 → 3.00 | 0.42 / 0.42 | 74.0% / 26.0% ($T^*_{\text{think}} = 4$/t) | \$0.6957 → \$0.5130 (3-t cum) | 26.26% | 785.5 + 662.0 = 1,447.5 ms/t | 100.00% / 100.00% |
| **Pooled Mean** ($N = 1{,}000$)§ | **466.7 → 384.9 (439.8→372.3/t)** | **17.52%** | **121.6 / 74.3 (134.7 wf; 125.1/t)** | **1.55 → 1.40** | **0.12 / 0.17** | **70.5% / 29.5%** | **\$1.7608 → \$1.0663** | **39.44%** | **858.5 + 630.9 = 1,489.4 ms** | **99.70% / 100.00%** |

†PromptVM Turn-1 Pass@1 / Post-Verification Wf shown (Baseline Turn-1 Pass@1 is 65.00% in A, 100.00% in B/C/E, 60.00% in D [92.50% Post-Retry], and 85.00% Pooled [98.50% Post-Retry]).
‡In routed workflows, Flash appends a `schema_audit` diagnostic trailer (208.2 tok in A vs. 61.8 tok SQL-only [\$0.0605/1k]; 40.4 tok in B; 34.3 tok in C); in D, Flash emits 94.0-tok unified diffs while cyclic Argon emits 434.2 tok.
§Cohort E reports per-turn (/t) and 3-turn cumulative (cum) tokens alongside 3-turn workflow cost and per-turn latency (1,447.5 ms/turn, 4,342.5 ms 3-turn total); Pooled Mean reports workflow totals alongside per-turn means.

structured instructions $I$ in *Schema-Safe LLMLingua + Stage 1 AST* still drops negative modal qualifiers `never`/`not` (81.20% Turn-1 / 94.60% Wf). To ensure a strictly fair comparison against standalone routers (FrugalGPT [21] and RouteLLM [22], calibrated on $N_{\text{cal}} = 250$ to target $Q^* = 0.95$), we evaluate RouteLLM both as a standard binary tier selector ($T_{\text{think}} = 0$, reaching 91.20% Turn-1 / 98.60% Wf) and augmented with access to the exact same reasoning-enabled Flash-Thinking action space ($\{(\text{Flash}, T_{\text{think}} \in \{0, 8, 16, 128\}), (\text{Argon}, 0)\}$): *RouteLLM + Flash-Thinking (Raw)* (\$1.1640/1k, 94.40% Turn-1 / 99.10% Wf) and *Compaction + RouteLLM + Flash-Thinking* (\$1.0980/1k, 96.80% Turn-1 / 99.40% Wf) improve over binary routing but still underperform PromptVM's closed-form AST-cyclicity-aware Eq. (26) allocation (\$1.0663/1k, 99.70% Turn-1 / 100.00% Wf) because text-embedding routers fail to detect structural schema/import cycles ($\delta_{\text{inst}} = 0.030$) on 3.2% of traces.

Finally, to verify external generalization at full benchmark scale with 95% bootstrap confidence intervals (and zero benchmark-specific router recalibration, where the AST classifier maps SQL tasks zero-shot to Cohort A [$\delta_{\text{A}} = 0.000, \hat{L}_{\text{out,A}} = 64$] and function-calling tasks zero-shot to Cohort B [$\delta_{\text{B}} = 0.015, \hat{L}_{\text{out,B}} = 96$]), we evaluated PromptVM across three public benchmarks: (1) on the **full BIRD-SQL Dev** benchmark [13] ($N = 1{,}534$ queries across all 11 databases, alongside our $n = 200$ stratified subset), Stages 1–3 on fixed Gemini 4.0 Argon achieved $40.8\% \pm 0.9\%$ input token compaction (614.2 → 363.6 tokens; 41.2% on $n = 200$), lifted execution accuracy from $68.4\% \rightarrow 69.1\% \pm 1.2\%$ ($68.5\% \rightarrow 69.0\%$ on $n = 200$), and reduced cost by 34.5%, while enabling Stage 5 Pareto routing preserved $69.1\% \pm 1.2\%$ execution accuracy and increased total cost savings to **60.9% ± 1.4%** (61.4% on $n = 200$); (2) on the **full Berkeley Function Calling Leaderboard (BFCL)** AST evaluation suite [2] (https://gorilla.cs.berkeley.edu/leaderboard.html, $N = 2{,}000$ test cases across simple, multiple, parallel, and parallel-multiple AST categories), Stages 1–3 on fixed Gemini 3.8 Flash achieved $38.1\% \pm 0.7\%$ token compaction (509.5 → 315.4 tokens; 38.4% on $n = 200$), improved AST accuracy from $94.6\% \rightarrow 95.1\% \pm 0.6\%$, and reduced cost by 36.4%, while enabling Stage 5 dynamic thinking allocation lifted AST accuracy to **95.5% ± 0.5%** with zero regressions; and (3) on the public **RouterBench** benchmark [25] ($N = 1{,}000$ stratified prompts across MMLU, GSM8K, MBPP, and MT-Bench mapped across Flash, Flash-Thinking, and Argon at target quality $Q^* = 0.95$), Stage 5 reduced total inference cost by **34.2% ± 1.1%** relative to FrugalGPT and **14.8% ± 0.8%** relative to RouteLLM while maintaining $95.3\% \pm 0.7\%$ Pass@1.

## VI. RELATED WORK

**Prompt and Tool-Schema Compression:** Prior prompt compression works such as LLMLingua [17], LongLLMLingua [18], and Selective Context [20] filter tokens with low surprisal under small language models, while LLMLingua-2 [19] introduces task-agnostic data distillation. However, token-level statistical filtering frequently prunes negative polarity constraints, quotation delimiters, and numerical bounds, causing functional failures in structured tool and enterprise database tasks (such as BIRD-SQL [13], Spider 2.0 enterprise SQL workflows, and BFCL [2]). Recent deterministic tool-schema compression compilers (such as TSCG) and empirical studies of structured prompt notations (evaluating token fragmentation across JSON, YAML, TOML, and XML representations) demonstrate the value of grammar-aware schema minification, while embedding-based tool-schema retrieval pipelines discard candidate API definitions based on query similarity (risking recall failures on multi-hop compositional tool chains). Going beyond format-level minification or lossy schema retrieval, PromptVM's Stage 1 performs lossless contract-preserving AST canonicalization and Unigram/BPE subword boundary-guarded `$defs`/`$ref` sub-tree hoisting across all active schemas while Stage 2 guarantees polarity-aware predicate retention.

**Attention Sinks, Modular KV-Cache Reuse, and Test-Time Compute Routing:** StreamingLLM [14] demonstrated that initial sequence tokens act as natural attention sinks in

TABLE III
SYSTEM COMPONENT ABLATION MATRIX AND BASELINE COMPARISON ($N = 1{,}000$, 5 SEEDS, MEDIAN SEED $s = 42$)

| System Configuration | In Tok | Out / $\bar{R}_k$ | Cost/1k | Latency | Turn-1 / Wf Pass† |
|---|---|---|---|---|---|
| Monolithic Baseline (Gemini 4.0 Argon) | 466.7 | 74.3 / 1.55 | \$1.7608 | 2,450.0 ms | 85.00% / 98.50% |
| Monolithic Baseline (Diff-Matched Cohort D, Argon) | 466.7 | 51.9 / 1.52 | \$1.4568 | 2,198.4 ms | 85.40% / 98.50% |
| Deterministic Schema Minification (RFC 8259 [8]) | 416.3 | 69.8 / 1.53 | \$1.6980 | 2,320.0 ms | 85.00% / 98.50% |
| Stage 1a Only (Key Sort + Union Fold + Guard, Argon) | 411.2 | 66.2 / 1.51 | \$1.6915 | 2,255.0 ms | 85.00% / 98.50% |
| Stage 1 Complete (1a + `$defs`/`$ref` Hoisting, Argon) | 408.5 | 65.4 / 1.51 | \$1.6881 | 2,240.0 ms | 85.00% / 98.50% |
| Stage 2 Only (Span-Strip + Invariant Pruning, Argon) | 440.6 | 62.1 / 1.50 | \$1.7282 | 2,190.0 ms | 82.10% / 95.80% |
| LLMLingua Compression (Full Prompt, Argon) [17] | 350.0 | 54.2 / 1.62 | \$1.4700 | 2,120.0 ms | 64.50% / 74.20% |
| LLMLingua-2 Distillation (Full Prompt, Argon) [19] | 342.0 | 53.0 / 1.58 | \$1.4480 | 2,080.0 ms | 71.80% / 82.40% |
| LLMLingua-2 (Matched Budget = 384.9 tok, Argon) [19] | 384.9 | 56.4 / 1.56 | \$1.5420 | 2,140.0 ms | 75.40% / 86.10% |
| Selective Context (Full Prompt, Argon) [20] | 373.3 | 55.8 / 1.60 | \$1.5280 | 2,190.0 ms | 68.20% / 78.50% |
| Schema-Safe LLMLingua ($I$ Only) + Stage 1 AST | 398.2 | 58.1 / 1.48 | \$1.5840 | 2,040.0 ms | 81.20% / 94.60% |
| FrugalGPT Cascade [21] | 466.7 | 72.0 / 1.49 | \$1.2840 | 2,150.0 ms | 88.40% / 97.40% |
| RouteLLM Preference Router (Raw, $T_{\text{think}} = 0$) [22] | 466.7 | 71.5 / 1.45 | \$1.2180 | 1,680.0 ms | 91.20% / 98.60% |
| RouteLLM + Flash-Thinking (Raw) [22] | 466.7 | 68.4 / 1.42 | \$1.1640 | 1,615.0 ms | 94.40% / 99.10% |
| Compaction + RouteLLM ($T_{\text{think}} = 0$) [22] | 384.9 | 64.2 / 1.43 | \$1.1120 | 1,540.0 ms | 94.80% / 98.80% |
| Compaction + RouteLLM + Flash-Thinking [22] | 384.9 | 63.0 / 1.41 | \$1.0980 | 1,512.0 ms | 96.80% / 99.40% |
| Compaction Only (on Gemini 4.0 Argon)§ | 384.9 | 42.7 / 1.40 | \$1.6586 | 1,980.0 ms | 100.00% / 100.00% |
| Compaction w/o Landmark Anchors (on Argon) | 382.4 | 42.1 / 1.48 | \$1.6550 | 1,970.0 ms | 82.10% / 95.80% |
| MINLP Routing Only (Raw Prompts) | 466.7 | 138.2 / 1.43 | \$1.3250 | 1,720.0 ms | 94.20% / 98.00% |
| **PromptVM Full Pipeline** | **384.9** | **134.7 / 1.40** | **\$1.0663** | **1,489.4 ms** | **99.70% / 100.00%** |

†Turn-1 Single-Call Pass@1 (Pre-Retry) / Post-Verification Workflow Completion Rate (Wf) under identical sandbox oracles.
§Output-normalized cost isolating pure input savings (−\$0.10225 at fixed 74.3 out tok); realized cost at 42.7 out tok and $\bar{R}_k = 1.40$ is \$1.2946.

autoregressive models. PromptVM establishes the complementary compile-time principle: synthetically injecting uppercase structural landmark anchors bounds background attention dispersion while co-locating invariant clauses to amplify retrieval-head attention. In runtime memory management, vLLM [10] and SGLang [11] manage PagedAttention block tables and RadixAttention prefix reuse, while modular KV-cache reuse frameworks such as Prompt Cache [23] and recent position-independent KV-cache block-padding systems (such as MiniPIC and MEPIC) precompute reusable attention blocks with decoupled positional encodings across document chunks; PromptVM provides the hardware-aligned complement that pads static prefixes to exact HBM2e page boundaries ($B \in \{16, 32\}$) with zero-initialized VMEM register masking and decoupled logical RoPE/sliding-window counters to eliminate copy-on-write boundary-block duplication across concurrent subagents. Finally, in model routing and test-time compute scaling, systems such as FrugalGPT [21], Hybrid LLM [24], RouteLLM [22], and unified routing evaluation suites such as RouterBench [25] and LLMRouterBench explore reactive post-generation cascading and preference-based tier selection, while recent difficulty-aware test-time compute scaling studies [26] and adaptive multi-tier test-time compute routers (such as BEST-Route) show that allocating variable reasoning compute often outperforms scaling model parameters. Rather than incurring two sequential model invocations via reactive POMDP cascades or sampling multiple parallel completions, PromptVM unifies these paradigms via a single-pass pre-dispatch closed-form discrete MINLP that jointly selects model serving tiers and integer test-time reasoning budgets ($T^*_{\text{think}}$) co-designed with upstream prompt compaction.

## VII. Conclusion

We presented the *Tokenomics Compiler* (`PromptVM`), a hardware-aware compilation and discrete-choice economic routing framework for enterprise LLM serving. By uniting tokenizer-aware AST transpilation, invariant density compaction, attention-sink landmark anchoring, server-side Pallas paged-cache boundary sealing, and discrete Pareto routing, our system slashes per-cohort enterprise inference costs by 26.26% to 88.79% (39.44% blended net savings; 26.81% against a diff-matched baseline), eliminates multi-branch copy-on-write boundary page duplication on Cloud TPU v6e, and achieves 99.70% strict Turn-1 Single-Call Pass@1 and 100.00% workflow completion while provably bounding attention dispersion.

*Limitations and Trade-offs:* First, compaction efficiency is bounded by the ratio of structured schemas and boilerplate to dense payload content; on raw source code refactoring (Cohort D), AST schema compression yielded a 1.11% token reduction, with cost savings driven primarily by Stage 5 dynamic reasoning allocation and unified-diff output structuring. Second, achieving $100.00\%$ post-verification workflow completion (lifting $99.70\%$ Turn-1 Pass@1 across the $0.30\%$ residual Cohort D failures) relies on deterministic execution-verification sandboxes (`go build`, Cloud Spanner AST parsers, JSON Schema validators) that add $482$–$714$ ms of verification latency; in open-ended natural language generation tasks lacking an executable oracle, workflow accuracy is governed by the pre-escalation Turn-1 rate ($99.70\%$). Third, the Stage 5 quality surface is calibrated on $N_{\text{cal}} = 250$ held-out traces; under distribution shift to unseen task domains, the router relies on its conservative fallback parameters ($\delta_{\text{default}} = 0.035, \hat{L}_{\text{out,default}} = 256, \hat{L}_{\text{sandbox,default}} = 600.0$ ms, $L_{\max,\text{default}} = 2{,}000$ ms), and Stage 2's static 64-token lexicon requires domain-specific extension via its configurable `custom_lexicon` interface when new domain-specific invariant markers arise. Fourth, Stages 4 and 5 exhibit an inverse infrastructure trade-off between managed cloud APIs and self-hosted accelerators: while Stages 1–3 and Stage 5 deploy transparently as a client-side SDK middleware on managed elastic cloud APIs (Vertex AI, where multi-tier routing across Flash and Argon incurs zero client memory overhead), Stage 4's interior page padding requires our custom server-side JAX/Pallas kernel on self-hosted Cloud TPU v6e clusters (as managed APIs would otherwise bill and attend to literal pad tokens); conversely, operating Stage 5's multi-tier routing on a fixed-capacity self-hosted TPU pod requires partitioning HBM across multiple model weights ($\{\text{Flash}, \text{Argon}\}$), trading per-tier KV-cache capacity against routing elasticity. Future work will extend AST transpilation to programming language syntax trees via Tree-sitter parsers and integrate multi-tier host DRAM-to-HBM speculative page prefetching.

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